How Fast Does the Inverse Walk Approximate a Random Permutation?

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September 14, 2025

Abstract

For a finite field $\mathbb{F}$ of size n, the (patched) inverse permutation $\mathrm{INV} : \mathbb{F} \to \mathbb{F}$ computes the inverse of x over $\mathbb{F}$ when $x \neq 0$ and outputs 0 when $x = 0$, and the ARK_K (AddRoundKey) permutation adds a fixed constant K to its input, i.e.,

$$\mathrm{INV}(x) = x^{n-2} \quad \text{and} \quad \mathrm{ARK}_K(x) = x + K \ .$$

We study the process of alternately applying the INV permutation followed by a random linear permutation ARK_K, which is a random walk over the alternating (or symmetric) group that we call the *inverse walk*.

We show matching upper and lower bounds on the number of rounds it takes for this process to approximate a random permutation over $\mathbb{F}$. We show that r rounds of the inverse walk over the field of size n with

$$r = \Theta\left(n \log n + n \log \frac{1}{\varepsilon}\right)$$

rounds generate a permutation that is ε-close (in total variation distance) to a uniformly random permutation of the appropriate sign. Our bound on r is optimal, up to a constant. In fact, we prove stronger lower bounds showing that the inverse walk needs at least $t = \Omega(n \log(1/\varepsilon))$ rounds to become ε-approximately 4-wise independent and at least $s = \Omega(n \log n)$ rounds to get to within $(1 - o_n(1))$ in total variation distance of a given 4-wise independent permutation.

Our result answers an open question from the work of Liu, Pelecanos, Tessaro, and Vaikuntanathan (CRYPTO 2023) by proving the t-wise independence of (a variant of) AES for t up to the square root of the field size, compared to the original result that only held for $t = 2$. It also constitutes a significant improvement on a result of Carlitz (Proc. American Mathematical Society, 1953) who showed a *reachability result*: namely, that every even permutation can be generated *eventually* by composing INV and ARK. We show a *tight convergence result*, namely a tight quantitative bound on the number of rounds to reach a random (even) permutation.

Our work brings to the forefront the view of block ciphers as random walks and uses novel combinatorial and analytic tools to study their pseudorandomness, both of which we hope will prove useful in the study of block ciphers.

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1 Introduction

The design and analysis of block ciphers such as the Advanced Encryption Standard (AES) [DR02] is a central topic in cryptography. On the one hand, despite extensive cryptanalysis, spanning a wide range of attacks including linear [MY92] and differential [BS91] cryptanalysis, higher-order [Lai94], truncated [Knu94] and impossible [Knu98] differential attacks, interpolation [JK97] and algebraic attacks [CP02], integral cryptanalysis [KW02], biclique attacks [BKR11], there has not been a single devastating attack thus far that undermines our confidence in AES. On the other hand, the situation is extremely unsatisfactory from a foundational perspective: indeed, it is not clear whether it is even possible to formulate a meaningful non-tautological computational hardness assumption that implies the security of AES within the classical framework of provable security. Given that AES is the integral part of the transport layer security (TLS) protocol that lies at the very foundations of secure internet communication, understanding its security is one of the most important questions in cryptography.

In this work, we continue the line of research that attempts to formally prove the security of block ciphers against *restricted* classes of attacks, with a focus on *substitution permutation networks* (SPNs), an important class of block ciphers that includes AES. The guiding principle of this line of study is to gradually expand the class of attacks we consider to include a large set of known cryptanalytic paradigms. In particular, we build on a recent pair of works by Liu, Pelecanos, Tessaro, and Vaikuntanathan (LPTV) [LTV21, LPTV23] who study the t-wise independence of SPNs, a statistical security property that prevents all t-input statistical attacks including differential and linear cryptanalysis (with $t = 2$) and higher order differential attacks (with larger t).[1]

Censored AES. The first of these works [LTV21] showed the 2-wise independence of the AES construction with many (over 9000) rounds. In an attempt to show comparable results with a lower number of rounds and show t-wise independence for large t, the second work [LPTV23] looked instead at SPNs with uniformly chosen *random* and secret S-boxes, and proved approximate t-wise independence (Definition 2.1) of a construction called AES*, first introduced by Baignères and Vaudenay [BV05], which differs from AES in that it uses such random S-boxes.

Theorem 1.1 ([LPTV23], Corollary 1)**.** *Assuming $t < 2^{(0.499-1/(4k))b}$, a $\Theta(k)$-round SPN* with maximal-branch-number linear mixing[2] is $2^{-\Theta(bk)}$-approximately t-wise independent.*

They also suggest a generic way to instantiate their results with the concrete AES S-box, i.e., the patched inverse permutation $\mathrm{INV} : x \mapsto x^{2^8-2}$ over the binary extension field $\mathbb{F}_{2^8}$. They observe in particular that a key-alternating cipher obtained by iterating INV, alternated with adding a random sub-key between each two sequential calls of INV, converges pretty quickly to being a *pairwise independent* permutation. We refer to this construction as the "INV KAC." Replacing the random S-box in their AES* result with the INV-KAC, they obtain in particular the following result, which they cast in terms of a construction they call "censored AES", which is essentially AES with some of the mixing layers removed. An example consequence of their result with concrete parameters is:

Theorem 1.2 ([LPTV23], Theorem 7)**.** *192-round censored AES is 2^{-128}-approximately pairwise independent.*

[1] We note here two caveats of their results: the first is that they assume the round keys are independent; and secondly, their proof works for many rounds of SPN/AES, although this has been improved in subsequent works.

[2] The branch number quantifies how well the linear mixing combines the k input blocks. For example, a trivial Linear Mixing that doesn't combine the k input blocks between them will have a very low branch number, and an SPN* with it will not achieve any pseudorandomness.

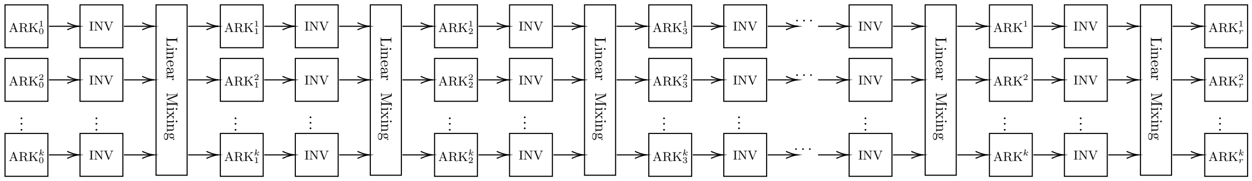

Figure 1: A schematic representation of the class of substitution permutation networks (SPN), which includes AES. The input to the block cipher is a string of bk bits, arranged into k blocks of b bits. We view the b bits of each block as an element of the field $\mathbb{F}_{2^b}$. The block cipher proceeds for r rounds, and in round i, each block j is shifted by a random secret key using the ARK_i^j operation (we are abusing notation for simplicity, since the subscript i in ARK_i does not refer to the secret key, but rather to the round). Next, we replace each block with its inverse in the field using the INV operation. Finally, the Linear Mixing step replaces the k blocks with linear combinations of their contents to "mix" them. Since the only step that is randomized is the ARK step, we perform it once more at the end of the cipher, for a total of $r + 1$ ARK operations.

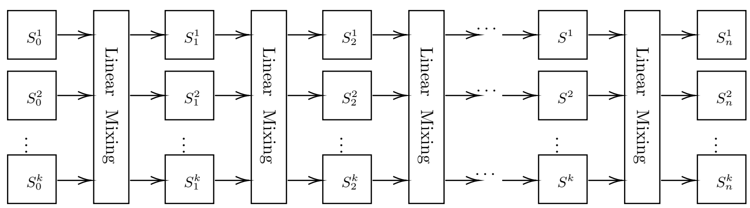

Figure 2: An example of an SPN with uniformly random and secret S-boxes, which [LPTV23] refer to as SPN*. Similar to the original SPN construction, each input is arranged in k elements of $\mathbb{F}_{2^b}$. Instead of successively applying the ARK and INV operations, an SPN* instead applies a uniformly random and secret permutation S_i^j to the j-th block of its input during round i. The Linear Mixing operation acts as before, by taking linear combinations of the blocks. Due to its structure, it is often easier to understand and prove theorems on how an SPN* scrambles its inputs, compared to an SPN.

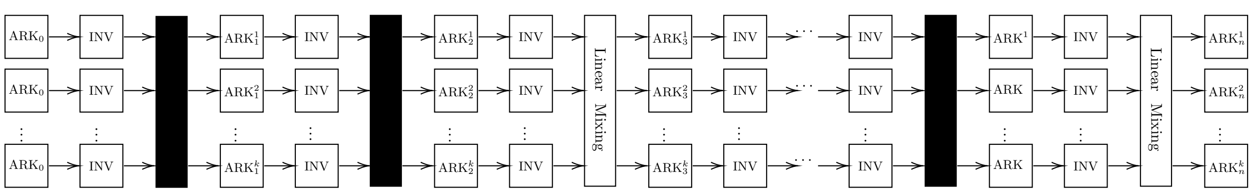

Figure 3: An example of censored AES. [LPTV23] suggest a generic way to instantiate their SPN* results using the censored AES. That is, they replace each random and secret permutation S_i^j in the SPN* construction with consecutive layers of ARK and INV (while censoring the Linear Mixing steps between them). In the figure above, the first 3 pairs of ARK and INV would correspond to the first layer of $S_0^1, \dots, S_0^k$ in an SPN*. The number of $(\mathrm{ARK}, \mathrm{INV})$ layers needed to approximate a uniformly random permutation S_i^j is related to the mixing time of the INV KAC random walk, which is the focus of this paper.

It is natural to conjecture that the actual AES cipher is not less secure than its censored counterpart, i.e., additional mixing layers only help, and so one can conjecture that the bound extends to 192-round AES, hence improving the bound of [LTV21] under this conjecture.

Their result however only applies to pairwise independence. This calls the question of whether the INV-KAC can be proved to be t-wise independent, in order to prove similar results for $t > 2$. In [LTV21], the authors sketch a proof for why this cipher needs at least linear in the size of the field many rounds to converge to a 4-wise independent permutation, but we note that this is not necessarily a major limitation in a context where the domain itself has small size, i.e, 256 elements. This argument relies on a similar idea as the interpolation attack by Jakobsen and Knudsen [JK97].

Our contributions. In this work, we prove matching lower and upper bounds on the number of rounds for the INV KAC to reach t-wise independence. We note here that since the cipher composes an AddRoundKey (henceforth ARK) operation with an INV operation, each round of the cipher generates a permutation with a fixed parity (the parity depends on the size of the field). Thus, we can only hope for the cipher to converge to the appropriate coset of the alternating group (as opposed to the symmetric group) which means that t has to be at most $n - 2$, for n being the size of our field. Here, and throughout the rest of this paper, we will quantify the convergence of distributions using the total variation distance: for probability distributions μ and ν on a finite state space Ω, the total variation distance between μ and ν is given by

$$\|\mu - \nu\|_{\mathrm{TV}} := \frac{1}{2}\sum_{\omega \in \Omega} |\mu(\omega) - \nu(\omega)|.$$

Theorem (Lower bounds, Theorems 4.1 and 4.2)**.** *An r-round INV KAC over a field of size n requires at least $r = \Omega(n \log(1/\varepsilon))$ rounds to become ε-approximately 4-wise independent and $r = \Omega(n \log n)$ rounds to get to within $(1 - o_n(1))$ in total variation distance of a given 4-wise independent permutation.*

Theorem (Upper bound, Theorem 3.1)**.** *An r-round* INV *KAC over the field of size n with*

$$r = O\left(n \log n + n \log \frac{1}{\varepsilon}\right)$$

rounds generates a permutation that is ε-close in total variation distance to an $(n-2)$-wise independent permutation (in fact, to a uniformly random permutation from the appropriate coset of the alternating group A_n).

We provide explicit constants in the formal version of our result, Theorem 3.1 (see also Remark 2.9), although we have not attempted to optimize them.

The proofs of our theorems view each round of the INV KAC as a step in a random walk over the symmetric group, starting from the identity permutation. In each step, you apply a random one of the n many permutations $\Pi_K(x) = \mathrm{INV}(x + K)$ where the addition is over the underlying field. Thus, a random walk of length r in this graph is equivalent to a composition of r many random permutations $\Pi_{K_1}, \Pi_{K_2}, \ldots, \Pi_{K_r}$, where the randomness is over the choice of the round keys $K_1, K_2, \ldots, K_r$. The problem then reduces to bounding the distance between the distribution of this random walk and the uniform distribution over the appropriate coset of A_n (here, n is the size of the field which, in the case of AES, is 2^8). We note that a prior version of our upper bound theorem first appeared in the appendix of [LPTV24] (which is the full version of [LPTV23] on eprint) with the polynomially worse bound $r \lesssim O(n^2 \log n)$, and only for fields of characteristic 2.

Having the upper bound, one can now extend the random S-box result in Theorem 3 of [LPTV23] to the concrete AES S-box. In particular, by following the same proof as in Theorem 7 of [LPTV23], but for larger t, we get the following corollary.

Corollary 1.3. *Assuming $t < 2^{(0.499-1/(4k))b}$, $\Theta\big(b2^b \cdot k\big)$-round censored SPN with k b-bit blocks, the AES S-box, and a maximal-branch-number linear mixing is $2^{-\Theta(kb)}$-approximately t-wise independent.*

The Censoring Conjecture. If one believes that the mixing layers are useful for AES to achieve pseudorandomness, then it is natural to expect that removing a large fraction of them should only hurt the convergence to t-wise independence. This leads us to conjecture that r-round AES is not less secure than r-round censored AES.

Conjecture 1.4 (Censoring conjecture)**.** *For any t, there exists a fixed constant $r_{\text{thresh}} > 0$, such that for any number of rounds $r > r_{\text{thresh}}$, r-round AES is ε-approximately t-wise independent if the same is true for r-round censored AES.*

We introduce a small round threshold r_{thresh} to allow the block cipher to enter a "generic" state, and avoid any non-typical behavior that may be happening during the first few rounds.

It is worth noting that a censoring result of a similar flavor has appeared in the Markov chain literature by Peres and Winkler [PW11]. Their setting is limited to a very specific family of Markov chains (Glauber dynamics in a monotone spin system, when the starting position is an extremal configuration) that do not include the behavior of a typical block cipher. Nevertheless, this censoring result has found many applications, and fully understanding how censoring affects the mixing time of a Markov chain remains an open problem.

A Mathematical Motivation. In 1963, Carlitz [Car53, Car63, Zie13] proved that the group of all permutations of $\mathbb{F}_q$ is generated by the permutations induced by degree-one polynomials and INV. Our result extends Carlitz's theorem in two ways. First, we show that if we restrict our attention to degree-one polynomials whose linear coefficient is 1, then we still generate at least the group of all even permutations[3]. Second, while Carlitz shows that every permutation can be reached via a composition of degree-one polynomials and INV, it does not tell us anything about how many steps it takes to do so: we provide a tight quantitative bound on the number of steps (operations) needed to reach a random permutation. We remark that the notion of Carlitz rank of a permutation, namely the number of "Carlitz steps" one needs to take to get the permutation, has been studied in the literature; see [AcMT09, Top14, IW18].

In reverse, the study of this mathematical problem brings to bear sophisticated techniques from both finite field theory (cf. [Car53, Car63] and followups) and Markov chain theory to the practical problem of proving block cipher security.

Related Work. We believe that the INV KAC is an important cipher design to study for two reasons. First, many substitution-permutation networks (SPNs) like the Advanced Encryption Standard (AES) [DR02] use the INV as their (non-linear) S-box permutation. Thus studying the INV "in isolation" may provide insight into its strengths and weaknesses. Additionally, our result shows that the INV KAC is a block cipher that we can understand almost exactly.

Furthermore, our analysis of the INV KAC, perhaps in conjunction with generalizations of Carlitz's result to general power maps [Sta98], may be useful in the analysis of other KACs, such as MiMC [AGR+16], which uses the cube permutation (over an appropriate finite field) instead of the patched inverse.

[3] In some cases this is the best we can do, e.g. over $\mathbb{F}_7$, since both the INV and ARK are even permutations. For other fields (including fields of characteristic 2), we can generate the entire symmetric group since we combine both even and odd permutations.

1.1 Our techniques

Optimal mixing times via comparison. In [LPTV24], an upper bound of the form $O(n^2 \log n \log(1/\varepsilon))$ on the mixing of the reversible INV KAC walk in the case when $\mathbb{F}_n$ has characteristic 2 is proved by bounding the *spectral gap* of reversible INV KAC using the classical comparison method (see, e.g.,[Sin92, DSC93b]). In our setting, as is the case for many local walks on the symmetric group, the functional inequality capturing the correct order of magnitude of the mixing time is presumably the *modified log-Sobolev inequality* ([Goe04]). However, bounding the modified log-Sobolev constant using the comparison method is extremely delicate and has only been accomplished recently in a very special case [TY24] (the same work [TY24] provides a general comparison theorem for the modified log-Sobolev constant; however, it is unclear how to construct a "flow" in our situation for which this theorem yields the desired bound). Instead, we exploit the fact that the reversible formulation of INV KAC (see Definition 2.4) is a symmetric random walk on a group. This allows us to use the comparison theorems of Diaconis and Saloff-Coste [DSC93a] to control the *entire spectrum* (as opposed to just the spectral gap) of reversible INV KAC using comparison theory, and hence allows us to effectively bound the ℓ_2-mixing time. From here, control of the total variation mixing time follows by Cauchy–Schwarz.

At a high level, the comparison method requires one to construct a multi-commodity flow between pairs of vertices that are connected in a Markov chain, by using paths over the P_n^{INV} edges (which correspond to composing $\mathrm{INV}$and ARK operations). Furthermore, this flow must have low congestion, that is, we would like the flow passing through every edge in P_n^{INV} to be roughly uniform. Given the strategy outlined in the previous paragraph, the main work in proving the upper bound is to compare the Dirichlet form of INV KAC with that of simpler Markov chains on the symmetric group for which the optimal ℓ^2 mixing time is known (or can otherwise be deduced) by constructing low congestion flows; in our case, these simpler Markov chains will ultimately be the well-studied random k-cycle walk $P_n^{k\text{-cyc}}$ (Definitions 2.6 and 2.7), where k is either 2 or 3 depending on the parity of n.

Auxiliary Markov chains. For fields of characteristic 2, we first compare INV KAC to the chain $P_n^{2\mathrm{cyc},0}$, in which a random step corresponds to transposing a uniformly random element with the element 0, and leaving all other elements intact. The key component in such a comparison is to show how to generate transpositions of the form $(0, i)$ for any non-zero i using ARK and INV. To do this, we extend the proof of Carlitz [Car63]. Next, we compare $P_n^{2\mathrm{cyc},0}$ to the well-studied random transpositions walk $P_n^{2\mathrm{cyc}}$ where each step is a random transposition of the form (i, j) that swaps i and j and leaves the rest of the domain intact [DS81, LY98, FOW18, Sal20]. Since the random transposition walk is periodic, we cannot use standard comparison lemmas, and so we prove a periodic comparison lemma.

For fields of odd characteristic, we compare the P_n^{INV} Markov chain to the chain $P_n^{3\mathrm{cyc,ap}}$, in which a random step corresponds to applying a 3-cycle chosen uniformly at random from a specific subset of all 3-cycles (that involve elements from an arithmetic progression). Then, we compare $P_n^{3\mathrm{cyc,ap}}$ to the well-studied Markov chain $P_n^{3\mathrm{cyc}}$ that applies a random 3-cycle at every step [Goe04, STY22]. We do this by showing how one can combine a constant number of these arithmetic-progression 3-cycles to generate an arbitrary 3-cycle over the n elements.

One may wonder whether there exists a common upper bound approach for fields of characteristic both odd and even. We remark that there is no hope of producing a 2-cycle in $\mathbb{F}_n$ if $n \equiv 3 \mod 4$ since both ARK and INV are then even permutations. Our method for producing 3-cycles does not immediately translate to even n, as we first produce an arithmetic progression of length 3, which has no corresponding notion in characteristic 2. Interestingly, the case $n \equiv 1 \mod 4$ can be handled by either argument. As our argument for odd characteristic provides a better constant, we will consider $n \equiv 1 \mod 4$ as a subcase of odd characteristic. However, in Appendix B, we write our proof for even characteristic to handle any field with a square root of -1.

Overview of the flow construction. Our comparison approach in the characteristic 2 case follows in the footsteps of [Car63] and shows how to perform a transposition. Our exact path construction turns out to be very different from Carlitz's, as the ARK operation does not implement all degree-one polynomials (as Carlitz requires) but is restricted to linear *shifts*. Instead, we show that, given ι satisfying $\iota^2 = -1$, we can implement the following set of transpositions

$$\left\{\left(u + v(uv+\iota)^{-1}, u + (v+1)(uv+u+\iota)^{-1}\right)\right\}_{u,v}$$

for most values of $u, v \in \mathbb{F}_n$, using the following 7 rounds (8 random keys) of the INV KAC:

$$\begin{aligned}&\mathrm{ARK}_{-u} \circ \mathrm{INV} \circ \mathrm{ARK}_{-u} \circ \mathrm{INV} \circ \mathrm{ARK}_{\iota v+\iota} \circ \mathrm{INV} \circ \mathrm{ARK}_{\iota} \circ \\ &\quad \circ \mathrm{INV} \circ \mathrm{ARK}_{\iota} \circ \mathrm{INV} \circ \mathrm{ARK}_{\iota v} \circ \mathrm{INV} \circ \mathrm{ARK}_{u} \circ \mathrm{INV} \circ \mathrm{ARK}_{u}\,.\end{aligned}$$

The resulting multi-commodity flow does not give an optimal lower bound for the Dirichlet form since the P_n^{INV} edges corresponding to ARK_{ι} suffer from higher congestion. To improve the congestion, we "spread out" the flow through these edges by demonstrating a randomized way to generate $\mathrm{ARK}_{\iota} \circ \mathrm{INV} \circ \mathrm{ARK}_{\iota}$ using 6 random keys that depend on the random variable $w \in \mathbb{F}_n$:

$$\begin{aligned}&\mathrm{ARK}_{-w+\iota} \circ \mathrm{INV} \circ \mathrm{ARK}_{-w^{-1}} \circ \mathrm{INV} \circ \mathrm{ARK}_{-w} \circ \mathrm{INV} \circ \\ &\quad \circ \mathrm{ARK}_{w^{-1}} \circ \mathrm{INV} \circ \mathrm{ARK}_{-w} \circ \mathrm{INV} \circ \mathrm{ARK}_{w^{-1}+\iota}\,.\end{aligned}$$

One additional tweak to the first and last round keys is required to obtain a low-congestion set of P_n^{INV} paths that transpose the element 0 with an almost uniformly random non-zero element i and complete the comparison with $P_n^{\mathrm{2cyc},0}$.

Additional challenges come up when we move to fields of odd characteristic. This is because when the size of the field n is a prime congruent to 3 modulo 4, then the INV operation transposes every element except $\{-1, 0, 1\}$. Thus it computes an even permutation, since it consists of $\frac{n-3}{2}$ transpositions. Moreover, the ARK_k operation is also an even permutation. ARK_0 is clearly even. Otherwise, if $n = p^b$ for p prime, then ARK_k consists of p^{b-1} disjoint p-cycles for $p^{b-1}(p-1)$ total cycles, which is even if n (and so p) is odd. We conclude that one cannot construct a transposition, which is an odd permutation, by composing INV and ARK. This is why our construction for the odd characteristic goes through 3-cycles, which are an even permutation. Our starting point is the observation that the following sequence of operations

$$\mathrm{ARK}_{-u-2v^{-1}} \circ \mathrm{INV} \circ \mathrm{ARK}\, v \circ \mathrm{INV} \circ \mathrm{ARK}_{-2v^{-1}} \circ \mathrm{INV} \circ \mathrm{ARK}_{v} \circ \mathrm{ARK}_{u}$$

generates the 3-cycle $(-u, -u-2v^{-1}, -u-v^{-1})$ for any u, and $v \neq 0$. This is precisely the subset of 3-cycles whose elements are terms of an arithmetic progression, as $(-u - v^{-1}) - (-u) = (-u - 2v^{-1}) - (-u - v^{-1})$. Furthermore, since every ARK operation in the above sequence has an almost uniformly random key, the resulting multi-commodity flow has low congestion and allows us to compare P_n^{INV} with $P_n^{\mathrm{3cyc,ap}}$.

Lower bound. For the lower bound, one may obtain $r = \Omega(n)$ rounds to become, say, $9/10$-approximately 4-wise independent by formalizing an argument of [Nyb93, LTV21], which relies on the following observation: for any collection of four inputs $X = \{x_1, x_2, x_3, x_4\} \subseteq \mathbb{F}$, the output of an r-round INV KAC walk with $r = o(n)$ on X coincides (with probability $1 - o_n(1)$) with the output on X of a function which is completely determined by its outputs on x_1, x_2, x_3. Since a 4-wise independent permutation is unlikely to satisfy this property, one may use this to distinguish between $o(n)$-round INV KAC and a 4-wise independent family.

For our optimal lower bound on total variation mixing (Theorem 4.2), we design a new estimator: instead of just checking whether the output of an r-round INV KAC agrees with a low-complexity function (as above) on a given set of 4 points, we instead *count* the number of sets of size 4 on which the two functions agree, and use this count to distinguish between an r-round INV KAC with $r < n \log n/10$ and a 4-wise independent permutation. The analysis of this algorithm is based on a coupon-collector type argument. The same ingredient which goes into the design and analysis of this estimator is also used to obtain the optimal lower bound of $\Omega(n \log(1/\varepsilon))$ for achieving ε-approximate 4-wise independence. We present details in section 4.

2 Preliminaries

For the entirety of this paper, our inputs and operations will be over a finite field $\mathbb{F}_n$, where n is a prime power p^b. For fields of odd characteristic, we use 2 to denote the sum of the multiplicative identity with itself. We denote by INV the inverse over the field that maps x to x^{n-2}. It holds that $\mathrm{INV}(x) \cdot x = 1$ for all non-zero x, and $\mathrm{INV}(0) = 0$. We will also define ARK_K to be the AddRoundKey operation with (secret) round key K that maps x to $x + K$. Moreover, we use the symbols $\gtrsim, \lesssim$ to compare two asymptotic quantities without specifying the constant factors.

2.1 The INV Key-Alternating Cipher

A Key-Alternating Cipher (KAC) is parameterized by a field size n, number of rounds r, and fixed permutation $P : \mathbb{F}_n \to \mathbb{F}_n$. A KAC is a family of functions indexed by $r + 1$ sub-keys $K_0, K_1, \ldots, K_r$, and defined recursively as follows:

$$F_P^{(0)}(x) = x + K_0$$

$$F^{(i)}_{P,K_0,\ldots,K_i}(x) = P\Big(F^{(i-1)}_{P,K_0,\ldots,K_{i-1}}(x)\Big) + K_i.$$

The family of functions is

$$\mathcal{F}_P = \{F^{(r)}_{P,K_0,\ldots,K_r}(x) \mid K_i \in \mathbb{F}_n\} .$$

One can also naturally extend this to have different permutations in each round. In this paper, we consider the INV KAC, for which we use the INV map over $\mathbb{F}_n$ as the fixed permutation P.

Definition 2.1 (Approximate t-wise independence)**.** *A probability measure μ on the symmetric group is* ε-approximately t-wise independent *if for all distinct points $(x_1, \ldots, x_t)$, the distribution of their images $(f(x_1), \ldots, f(x_t))_{f\sim\mu}$, is at most ε away in total variation distance from a uniformly random t-tuple of distinct elements of $\mathbb{F}_n$.*

We say that a KAC $\mathcal{F}_P$ is ε-approximately t-wise independent if the uniform measure over $\mathcal{F}_P$ is ε-approximately t-wise independent.

It follows that a KAC that is ε-approximately t-wise independent is also ε-approximately to t'-wise independent for any $t' < t$. Moreover, the uniform distribution on even (or odd) permutations of $\mathbb{F}_n$ is $(n-2)$-wise independent.

2.2 The INV KAC as a Markov Chain

Let Π be the transition matrix of an ergodic Markov chain over a finite state space Ω, and let π denote its stationary distribution. We use E to refer to the set of edges of the underlying graph, that is $E = \{(x, y) : \Pi(x, y) > 0\}$. We identify a Markov chain with its transition matrix, so we will often say that Π is both the transition matrix for a Markov chain and also the Markov chain itself.

Definition 2.2 (Reversible Markov chain)**.** *A Markov chain* Π *is* reversible *with respect to the distribution* π *if for all* $x, y \in \Omega$*,*

$$\pi(x)\Pi(x, y) = \pi(y)\Pi(y, x).$$

This symmetry property implies that the operator $\Pi : \ell^2(\pi) \to \ell^2(\pi)$ is self-adjoint, thereby permitting spectral analysis.

Definition 2.3 (Mixing time)**.** *The* ε*-mixing time of an ergodic Markov chain* Π *is defined as the minimum number of steps* t *such that any random walk of length* t *is* ε*-close to the stationary distribution in total variation distance:*

$$\tau_\varepsilon(\Pi) = \min_{t \geq 0} \left\{ t : \max_{x \in \Omega} \|\delta_x \Pi^t - \pi\|_{\mathrm{TV}} \leq \varepsilon \right\},$$

where δ_x *denotes the point mass supported at* x*.*

To study the t-wise independence of the INV KAC cipher, we will model its execution as a random walk over the symmetric group of permutations over $\mathbb{F}_n$. Even though generating a truly random permutation may initially seem too strong for t-wise independence, we will see (perhaps surprisingly) that the number of rounds required to reach 4-wise independence and $(n-2)$-wise independence are close (up to constant factors). Thus considering convergence to the uniform distribution makes our analysis more convenient.

A first idea is to consider our block cipher to be a random walk over $\mathfrak{S}_n$, the symmetric group of permutations of $\mathbb{F}_n$, where every step of the walk applies first the ARK_K operation with a random key K from $\mathbb{F}_n$, and then the INV operation. The main issue of representing the block cipher this way is that the underlying Markov chain is not reversible with respect to the appropriate distribution, and thus it is harder to apply spectral machinery to analyze the mixing time of Markov chains.

We will instead use the following reversible Markov chain to represent our cipher:

Definition 2.4 (INV KAC Markov chain)**.** *The chain* P_n^{INV} *on the symmetric group* $\mathfrak{S}_n$ *has the following transition matrix. Given the current permutation* σ_t*, one step in this Markov chain corresponds to drawing uniformly and independently two random keys* K_1, K_2 *from* $\mathbb{F}_n$*, and setting*

$$\sigma_{t+1} = \mathrm{ARK}_{K_2} \circ \mathrm{INV} \circ \mathrm{ARK}_{K_1} \circ \sigma_t.$$

Note that the degree of the underlying graph has increased from n to n^2. It is not hard to observe that this transformation does not introduce any parallel edges to our Markov chain.

Lemma 2.5. *The Markov chain* P_n^{INV} *does not have any parallel edges for* $n \geq 5$*. That is, the following holds*

$$\mathrm{ARK}_i \circ \mathrm{INV} \circ \mathrm{ARK}_j = \mathrm{ARK}_k \circ \mathrm{INV} \circ \mathrm{ARK}_\ell,$$

only when $(i, j) = (k, \ell)$*.*

Proof. First, observe that if $i = k$, then the statement is true. Indeed, we can apply the permutation $\mathrm{INV} \circ \mathrm{ARK}_{-i}$ to both sides and obtain

$$\mathrm{ARK}_j = \mathrm{ARK}_\ell \implies j = \ell.$$

Similarly, if $j = \ell$, then the statement also holds. Hence, let us consider the case when both $i \neq k$, and $j \neq \ell$. For all $n - 2$ values of x such that $y = \sigma(x) \notin \{-j, -\ell\}$, x is mapped under the two permutations to equal values:

$$i + \frac{1}{y + j} = k + \frac{1}{y + \ell}$$

Therefore,

$$\begin{aligned}
&\frac{iy+ij+1}{y+j} = \frac{ky+k\ell+1}{y+\ell} \\
\implies\ & iy^2+(ij+1+i\ell)y+ij\ell+\ell = ky^2+(k\ell+1+kj)y+jk\ell+j \\
\implies\ & (i-k)y^2+((i-k)j+(i-k)\ell)y+(i-k)j\ell+(\ell-j)=0.
\end{aligned}$$

For the above equality to be true for more than 2 values of y, it must hold that $i=k$. This concludes the proof for $n \geq 5$. $\square$

2.3 Random k-cycle walks

We will bound the mixing time of the INV KAC Markov chain by comparing it to two well-studied Markov chains. In the first case, when $\mathbb{F}_n$ is of characteristic 2, we will use the random transposition Markov chain.

Definition 2.6 (Random transposition Markov chain)**.** *The chain P_n^{2cyc} on the symmetric group $\mathfrak{S}_n$ has the following transition matrix. Given the current permutation σ_t, one step in this Markov chain corresponds to drawing uniformly a random transposition (i,j) from $\mathfrak{S}_n$, and setting*

$$\sigma_{t+1} = (i,j) \circ \sigma_t.$$

In the second case, when $\mathbb{F}_n$ is of odd characteristic, we will use the random 3-cycle Markov chain.

Definition 2.7 (Random 3-cycle Markov chain)**.** *The chain P_n^{3cyc} on the alternating group A_n has the following transition matrix. Given the current permutation σ_t, one step in this Markov chain corresponds to drawing uniformly a random 3-cycle (i,j,k) from A_n, and setting*

$$\sigma_{t+1} = (i,j,k) \circ \sigma_t.$$

The ℓ^2-mixing time of both walks are well-understood [DS81, Hou16]. In each case, the correct asymptotic bound on the total variation mixing time follows by combining the below bound on the ℓ_2-mixing time with Cauchy-Schwarz [DS81].

Theorem 2.8 ([Hou16, Proof of Theorem 2])**.** *There exists an absolute constant $C_{2.8} > 0$ such that the following is true. Let $t \geq \frac{n}{k}(\log n + C_{2.8} R)$. Let π_ℓ be the uniform distribution on A_n if ℓ is even and on $\mathfrak{S}_n \setminus A_n$ if ℓ is odd. Then*

$$\|\delta_{\mathrm{id}}(P_n^{k\text{-cyc}})^t - \pi_{(k-1)t}\|_2^2 \leq \frac{e^{-R}}{|\mathfrak{S}_n|}.$$

Remark 2.9. *In this work, we will only need the above result for $k=2$ and $k=3$. For $k=2$, very good explicit bounds on $C_{2.8}$ are known [SCZ08, Theorem 1.1] and one can use similar techniques to obtain explicit bounds for the case $k=3$ as well. In order to keep this paper to a reasonable length (and since this would anyway be outside the scope of our work), we have chosen to present our main result Theorem 3.1 in terms of this constant $C_{2.8}$.*

2.4 The comparison method

Below we sketch the "comparison using multicommodity flows" method of Sinclair, Diaconis and Saloff-Coste [Sin92, DSC93b]. Let P^{ref} and P^{tar} be two reversible Markov chains on the same ground set Ω, with stationary distributions $\pi^{\mathrm{ref}}, \pi^{\mathrm{tar}}$ and edge sets $E^{\mathrm{ref}}, E^{\mathrm{tar}}$ respectively. We will think of P^{ref} as the "reference" chain for which we have somehow obtained estimates for the mixing time. Our goal is to leverage this in order to obtain estimates for the mixing time of P^{tar}. This will be accomplished by comparing the corresponding Dirichlet forms.

Definition 2.10 (Dirichlet form)**.** *Let Π be a Markov chain on Ω which is reversible with respect to π. Let $f, g : \Omega \to \mathbb{R}$ be real-valued functions. The* Dirichlet form *with respect to Π and π is*

$$\mathcal{E}_\pi^\Pi(f, g) = \frac{1}{2} \sum_{x,y \in \Omega} (f(x) - f(y))(g(x) - g(y))\pi(x)\Pi(x, y).$$

The following result of Diaconis and Saloff-Coste [DSC93a] allows us to compare ℓ^2-distances via a comparison of Dirichlet forms. In the statement below, we say that a probability measure μ on a group G is *symmetric* if

$$\mu(g) = \mu(g^{-1}) \qquad \forall g \in G.$$

Lemma 2.11 ([DSC93a, Lemma 5 and Remark])**.** *Let G be a finite group with symmetric probability measures $p^{\text{tar}}, p^{\text{ref}}$. Let $P^{\text{tar}} = (p^{\text{tar}}(xy^{-1}))_{x,y \in G}$ be the kernel of the random walk where each step is a sample from p^{tar}, and similarly let $P^{\text{ref}} = (p^{\text{ref}}(xy^{-1}))_{x,y \in G}$.*

If $\mathcal{E}^{\text{ref}}(f, f) \le A\, \mathcal{E}^{\text{tar}}(f, f)$ and P^{tar} has no negative eigenvalues, then

$$\|\delta_{\text{id}}(P^{\text{tar}})^t - \pi\|_2^2 \le e^{-t/A} + \|\delta_{\text{id}}(P^{\text{ref}})^{t/2A} - \pi\|_2^2,$$

where π is the uniform distribution on G and δ_{id} is the point mass at the identity element.

To deal with some periodicity issues, we will also need the following comparison result that holds for alternating walks. We defer the proof to Appendix A as it is derived very similarly to methods in [DSC93a].

Lemma 2.12. *Let G be a finite group with a subgroup H of index 2. Let $p^{\text{tar}}, p^{\text{ref}}$ be two symmetric probability measures supported on $G \setminus H$. Let $P^{\text{tar}} = (p^{\text{tar}}(xy^{-1}))_{x,y \in G}$ be the kernel of the random walk where each step is a sample from p^{tar}, and similarly $P^{\text{ref}} = (p^{\text{ref}}(xy^{-1}))_{x,y \in G}$.*

Let $A \in \mathbb{N}$. If $\mathcal{E}^{\text{ref}}(f, f) \le A\, \mathcal{E}^{\text{tar}}(f, f)$, then

$$\|\delta_{\text{id}}(P^{\text{tar}})^t - \pi_t\|_2^2 \le e^{-t/A} + \|\delta_{\text{id}}(P^{\text{ref}})^{\lfloor t/2A \rfloor} - \pi_{\lfloor t/2A \rfloor}\|_2^2,$$

where π_ℓ is the alternating measure that is uniform on H for ℓ even and uniform on $G \setminus H$ for ℓ odd.

Given the above results, our goal is to bound the Dirichlet form of P^{tar} by that of P^{ref}. Define a path γ_{xy} for $(x, y) \in E^{\text{ref}}$ to be a sequence of steps

$$(x = a_0, a_1, \ldots, a_k = y)$$

for which $(a_i, a_{i+1}) \in E^{\text{tar}}$. We say that such a path has length $|\gamma_{xy}| = k$. Let $\mathcal{P}_{xy}$ be the set of all simple paths connecting x to y. Also let $\mathcal{P} = \cup_{(x,y) \in E^{\text{ref}}} \mathcal{P}_{xy}$ be the union of all such paths. For $(a, b) \in E^{\text{tar}}$, let $\mathcal{P}(a, b) = \{\gamma \in \mathcal{P} \mid (a, b) \in \gamma\}$. That is, $\mathcal{P}(a, b)$ contains all paths that use the edge (a, b) of the target graph.

Definition 2.13 (Flow)**.** *A function $F : \mathcal{P} \to \mathbb{R}_{\ge 0}$ is called a $(P^{\text{tar}}, P^{\text{ref}})$-flow if*

$$\sum_{\gamma \in \mathcal{P}_{xy}} F(\gamma) = P^{\text{ref}}(x, y)\pi^{\text{ref}}(x).$$

The next result allows us to obtain a comparison of Dirichlet forms by constructing a suitable flow.

Theorem 2.14 ([DSC93b, Theorem 2.3])**.** *Let P^{tar}, P^{ref} be Markov chains on a finite set Ω reversible with respect to $\pi^{\text{tar}}, \pi^{\text{ref}}$ respectively. For any $(P^{\text{tar}}, P^{\text{ref}})$-flow F, the Dirichlet forms satisfy*

$$\mathcal{E}^{\text{ref}}(f, f) \le A(F) \cdot \mathcal{E}^{\text{tar}}(f, f)$$

with

$$A(F) = \max_{(a,b) \in E^{\text{tar}}} \left\{ \frac{1}{\pi^{\text{tar}}(a)P^{\text{tar}}(a, b)} \sum_{\gamma \in \mathcal{P}(a,b)} |\gamma| \cdot F(\gamma) \right\}.$$

Representing edges and paths. To employ the comparison method, we will need a way to specify paths in the P_n^{INV} Markov chain. Since each edge of P_n^{INV} is determined by two keys r_1, r_2, we will use the notation $[[r_1, r_2]]$ to denote the edge $\mathrm{ARK}_{r_2} \circ \mathrm{INV} \circ \mathrm{ARK}_{r_1}$. The starting vertex will be specified separately.

Whenever we need to describe a longer path of P_n^{INV}, we will write it as a tuple of double square brackets, by specifying the first edge first and so on, e.g. $([[r_1, r_2]], [[q_1, q_2]])$[4]. Since ARK operations form a subgroup, this path is also equal to

$$\begin{aligned}&(\mathrm{ARK}_{q_2} \circ \mathrm{INV} \circ \mathrm{ARK}_{q_1}) \circ (\mathrm{ARK}_{r_2} \circ \mathrm{INV} \circ \mathrm{ARK}_{r_1})\\ &= \mathrm{ARK}_{q_2} \circ \mathrm{INV} \circ \mathrm{ARK}_{q_1+r_2} \circ \mathrm{INV} \circ \mathrm{ARK}_{r_1}\,.\end{aligned}$$

We will also use the notation $[[r_1, r_2+q_1, q_2]]$ to describe the above path in P_n^{INV}. In general, we will extend the double square bracket notation to mean:

$$[[r_1, r_2, \ldots, r_k]] = \mathrm{ARK}_{r_k} \circ \mathrm{INV} \circ \cdots \circ \mathrm{ARK}_{r_2} \circ \mathrm{INV} \circ \mathrm{ARK}_{r_1}\,.$$

3 Upper Bound

In this section, we formally prove our claim that a random S-box over $\mathbb{F}_n$ can be approximated via the sequential composition of alternating AddRoundKey and INV S-box operations. The following is our main result.

Theorem 3.1. *Let $n \geq 5$. The* INV *KAC over the field of size n with $r \geq M \cdot n(\log n + C_{2.8} \log \frac{1}{\varepsilon})$ rounds generates a permutation that is ε-close to a uniformly random permutation from the appropriate coset of A_n. If $n \equiv 3 \mod 4$, then the coset is always A_n; if $n \not\equiv 3 \mod 4$, then the coset is A_n for r even and $\mathfrak{S}_n \setminus A_n$ for r odd.*

The constant M can be taken to be 1665 is n is even, 3888 if n is a power of 3, and 1296 if n has any other base.

We prove this separately depending on the parity of n in the following subsections.

3.1 Fields of even characteristic

The main work in this section is proving the following comparison inequality.

Lemma 3.2. *Let n be a power of 2. Then,*

$$\mathcal{E}^{\mathrm{2cyc}}(f, f) \leq 1665\, \mathcal{E}^{\mathrm{INV}}(f, f).$$

Proof. Our comparison method proceeds in two steps. We introduce an intermediate chain $P_n^{\mathrm{2cyc},0}$, a slight variant of the random transposition chain, and compare the Dirichlet form of P_n^{INV} with it. Then we compare the Dirichlet form of $P_n^{\mathrm{2cyc},0}$ to that of the random transposition chain P_n^{2cyc}. In the following two lemmas, a flow argument shows

$$\mathcal{E}^{\mathrm{2cyc}}(f, f) \underset{\text{Lemma 3.4}}{\leq} 9\, \mathcal{E}^{\mathrm{2cyc},0}(f, f) \underset{\text{Lemma 3.5}}{\leq} 9 \cdot 185\, \mathcal{E}^{\mathrm{INV}}(f, f). \qquad \square$$

[4]If we are writing the edges as permutations, we write the permutations corresponding to the edges from right-to-left.

Proof of Theorem 3.1 for even characteristic. We may compute

$$\begin{aligned}
\|\delta_{\mathrm{id}}(P_n^{\mathrm{INV}})^t - \pi_t\|_{\mathrm{TV}} &= \frac{1}{2}\|\delta_{\mathrm{id}}(P_n^{\mathrm{INV}})^t - \pi_t\|_1 \le \frac{1}{2}|\mathfrak{S}_n|^{1/2}\|\delta_{\mathrm{id}}P^t - \pi_t\|_2 \\
&\le \frac{|\mathfrak{S}_n|^{1/2}}{2}\Big(\exp(-t/1665) + \|\delta_{\mathrm{id}}(P^{2\mathrm{cyc}})^{\lfloor t/3330\rfloor} - \pi_{\lfloor t/3330\rfloor}\|_2^2\Big)^{1/2} \\
&\le \frac{1}{2}\left(|\mathfrak{S}_n|\exp\left(\frac{-t}{1665}\right)\right)^{1/2} + \frac{1}{2}\Big(|\mathfrak{S}_n|\cdot\|\delta_{\mathrm{id}}(P^{2\mathrm{cyc}})^{\lfloor t/3330\rfloor} - \pi_{\lfloor t/3330\rfloor}\|_2^2\Big)^{1/2}.
\end{aligned} \tag{1}$$

The first inequality is Cauchy–Schwarz; the second line uses Lemma 2.12; and the last is the numeric inequality $\sqrt{x+y} \le \sqrt{x} + \sqrt{y}$. Using Stirling's approximation for the first term and Theorem 2.8 for the second term, it follows that for $t \ge 1665\, n(\log n + C_{2.8}\log(1/\varepsilon))$, the right hand side is at most ε, as claimed. □

We have now reduced the problem to producing the two promised flows.

Definition 3.3 (Transposition with fixed element)**.** *The chain $P_n^{2\mathrm{cyc},0}$ on the symmetric group $\mathfrak{S}_n$ has the following transition matrix. Given the current permutation σ_t, one step in this Markov chain corresponds to drawing a uniformly random non-zero element i from $[n]$ and setting*

$$\sigma_{t+1} = (0,i)\circ\sigma_t.$$

First, we compare the Dirichlet forms of $P_n^{2\mathrm{cyc}}$ and $P_n^{2\mathrm{cyc},0}$.

Lemma 3.4. *The Dirichlet form of $P_n^{2\mathrm{cyc},0}$ satisfies*

$$\mathcal{E}^{2\mathrm{cyc}}(f,f) \le 9\,\mathcal{E}^{2\mathrm{cyc},0}(f,f).$$

Proof. We will divide the flow among either one or two paths for all edges $(x,y)\in E_n^{2\mathrm{cyc}}$. Let $y=(i,j)\circ x$ for some $i\ne j$. Then we will set $\mathcal{P}_{xy} = \{\gamma_{xy}\}$, where γ_{xy} is chosen according to the following two cases:

1. One of i,j is equal to zero. Then (i,j) is also in $E_n^{2\mathrm{cyc},0}$ and so we put all the flow along the one-step path
$$\gamma_{xy} = (x, (i,j)\circ x = y).$$
The flow is $F(\gamma_{xy}) = P_n^{2\mathrm{cyc}}(x,y)\cdot\pi(x) = \frac{1}{n!\binom{n}{2}} = \frac{2}{n(n-1)\cdot n!}$.

2. Both i,j are non-zero. Then we divide the flow along two paths
$$\begin{aligned}
\gamma_{xy} &= (x, \underbrace{(0,i)\circ x}_{a_1}, \underbrace{(0,j)\circ a_1}_{a_2}, (0,i)\circ a_2 = y), \\
\gamma'_{xy} &= (x, \underbrace{(0,j)\circ x}_{b_1}, \underbrace{(0,i)\circ b_1}_{b_2}, (0,j)\circ b_2 = y).
\end{aligned}$$
We equally divide the flow, so $F(\gamma_{xy}) = F(\gamma'_{xy}) = \frac{1}{n(n-1)\cdot n!}$.

The comparison constant we get is

$$
\begin{aligned}
A(F) &= \max_{(a,b)\in E^{2\mathrm{cyc},0}} \left\{ \frac{1}{\pi(a)\cdot P_n^{2\mathrm{cyc},0}(a,b)} \sum_{\gamma\in\mathcal{P}(a,b)} |\gamma|\cdot F(\gamma) \right\} \\
&= \max_{(a,b)\in E^{2\mathrm{cyc},0}} \left\{ n!(n-1)\cdot\left(\frac{2}{n(n-1)n!} + 3(n-2)\cdot 3\cdot\frac{1}{n(n-1)n!}\right)\right\} \\
&= \frac{2}{n} + 9\cdot\frac{n-2}{n} \leq 9.
\end{aligned}
$$

We used the fact that γ can occur in at most one path of length 1 and at most $3(n-2)$ paths of length 3. The first fact is obvious. The second follows as the edge (a,b) specifies a unique ℓ such that $b=(0,\ell)\circ a$. Consider a length-3 path γ_{xy} that uses edge (a,b), where $y=(i,j)\circ x$. This edge can be one of 3 edges of γ_{xy}, and the position of (a,b) in the path specifies one of i,j to be equal to ℓ. Thus the remaining variable has $n-2$ possible values (all except 0 and ℓ). $\square$

Next, we compare the Dirichlet forms of $P_n^{2\mathrm{cyc},0}$ and P_n^{INV}.

Lemma 3.5. *Let $n=2^b\geq 8$. The Dirichlet form of P_n^{INV} satisfies*

$$\mathcal{E}^{2\mathrm{cyc},0}(f,f)\leq 185\cdot\mathcal{E}^{\mathrm{INV}}(f,f).$$

We will need the following lemma, which provides a way to generate the transposition $(0,\frac{1}{(uv+1)(uv+u+1)})$ by combining INV and ARK operations. Its proof is deferred to Appendix B.

Lemma 3.6. *Let $n=2^b$. For any $u,v,w\in\mathbb{F}_n$ such that $uv\neq 1$, $uv+u\neq 1$, and $u\neq 0$, we can generate the transposition $\left(0,\frac{1}{(uv+1)(uv+u+1)}\right)$ using the following sequence $\psi_{\mathrm{even}}(u,v,w)$ of* ARK *and* INV *operations*

$$
\begin{aligned}
\psi_{\mathrm{even}}(u,v,w) = \Bigg[\Bigg[&\frac{v}{uv+1},\ u,\ v,\ \mathrm{INV}(w)+1,\ w,\\
&\mathrm{INV}(w),\ w,\ \mathrm{INV}(w),\ w+1,\ v+1,\ u,\ \frac{v}{uv+1}\Bigg]\Bigg].
\end{aligned}
$$

Using Lemma 3.6, we can construct paths that connect adjacent vertices of $P_n^{2\mathrm{cyc},0}$ using edges of P_n^{INV}.

Corollary 3.7. *Let $n=2^b$. For any $u,v,w\in\mathbb{F}_n$ such that $uv\neq 1$, $uv+u\neq 1$, and $u\neq 0$, and any $r_1,\dots,r_{10}\in\mathbb{F}_n$ we can generate the transposition $\left(0,\frac{1}{(uv+1)(uv+u+1)}\right)$ using the following path $\phi_{\mathrm{even}}(u,v,w,r_1,\dots,r_{10})$ of 11 edges of P_n^{INV}:*

$$
\begin{aligned}
\phi_{\mathrm{even}}(u,v,w,r_1,\dots,r_{10}) = \Bigg(&\Bigg[\Bigg[\frac{v}{uv+1},\ r_1\Bigg]\Bigg],\ [[u+r_1,\ r_2]],\ [[v+r_2,\ r_3]],\\
&[[\mathrm{INV}(w)+1+r_3,\ r_4]],\ [[w+r_4,\ r_5]],\ [[\mathrm{INV}(w)+r_5,\ r_6]],\ [[w+r_6,\ r_7]],\\
&[[\mathrm{INV}(w)+r_7,\ r_8]],\ [[w+1+r_8,\ r_9]],\ [[v+1+r_9,\ r_{10}]],\ \Bigg[\Bigg[u+r_{10},\ \frac{v}{uv+1}\Bigg]\Bigg]\Bigg).
\end{aligned}
$$

Proof of Lemma 3.5. Towards applying the Comparison Theorem (Theorem 2.14), we will only assign a non-zero amount of flow to the ϕ_{even} paths defined in Corollary 3.7. For any s, let

$$\Phi(s)=\{\phi_{\mathrm{even}}(u,v,w,r_1,\dots,r_{10})\mid (0,s)=\psi_{\mathrm{even}}(u,v,w), uv\neq 1, uv+u\neq 1, u\neq 0\}.$$

By Lemma C.1, for any $s \in \mathbb{F}_n$, the number of choices u, v, w satisfying the conditions for $\Phi(s)$ is $n(n-2)$. Note that $r_1, \ldots, r_{10}$ are completely free elements of $\mathbb{F}_n$, so for all $s \in \mathbb{F}_n$, $|\Phi(s)| = n^{11}(n-2)$.

For any $y = (0, s) \circ x$, all our paths will consist of starting at x and following some sequence of steps in $\Phi(s)$. We will equally distribute the flow among all these paths. Thus for any γ_{xy} with positive flow, we get

$$F(\gamma_{xy}) = \frac{1}{|\Phi(s)|} \cdot P_n^{\text{2cyc},0}(x, y) \cdot \pi(x) = \frac{1}{n^{11}(n-2)(n-1)n!}.$$

We are now ready to compute the comparison constant for this flow.

$$\begin{aligned}
A(F) &= \max_{(a,b) \in E_n^{\text{INV}}} \left\{ \frac{1}{\pi^{\text{INV}}(a) P_n^{\text{INV}}(a, b)} \sum_{\gamma \in \mathcal{P}(a,b)} |\gamma| \cdot F(\gamma) \right\} \\
&= \max_{(a,b) \in E_n^{\text{INV}}} \left\{ n! \cdot n^2 \cdot |\mathcal{P}(a, b)| \cdot 11 \cdot \frac{1}{n^{11}(n-2)(n-1)n!} \right\} \\
&= \frac{11}{n^9(n-2)(n-1)} \max_{(a,b) \in E_n^{\text{INV}}} |\mathcal{P}(a, b)| \\
&\leq \frac{11}{n^9(n-2)(n-1)} \cdot 11n^{11} = 121 \cdot \frac{n^2}{(n-2)(n-1)} < 185.
\end{aligned}$$

The last inequality holds for $n \geq 8$. We used the fact that $|\mathcal{P}(a, b)| \leq 11n^{11}$. This is because P_n^{INV} has no parallel edges (Lemma 2.5), and thus the edge (a, b) fully specifies a unique permutation of the form $[[k, \ell]]$.

Now consider a path $\gamma = \phi_{\text{even}}(u', v', w', r'_1, r'_2, \ldots, r'_{10})$ that uses edge $[[k, \ell]]$. This edge can be any one of 11 edges of γ; let's say that it is the i^{th} edge, for $i \in \{1, 2, \ldots, 11\}$. Every value of i implies two equations that the set of variables $\{u', v', w', r'_1, r'_2, \ldots, r'_{10}\}$ must satisfy. This restricts 2 of the 13 degrees of freedom; thus, we can have at most n^{11} such paths, since all paths are linearly dependent on the variables $(u', v', w', r'_1, r'_2, \ldots, r'_{10})$, or their inverses. $\square$

3.2 Fields of odd characteristic

Let $n = p^b$ be an odd prime power. In this case, ARK_k (for $k \neq 0$) consists of p^{b-1} disjoint p-cycles for $p^{b-1}(p-1)$ total transpositions, which is even. Clearly ARK_0 is even as well. However, the parity of INV depends on the value of n mod 4. The operation INV has three fixed points, so INV consists of $(n-3)/2$ transpositions and hence is even if and only if $n \equiv 3 \mod 4$. Using this, it is readily seen that if $n \equiv 3 \mod 4$, then INV KAC converges to the uniform distribution on the alternating group A_n while if $n \equiv 1 \mod 4$, then INV KAC converges to the uniform distribution on the appropriate coset (depending on the number of steps) of A_n. To handle both cases in a unified manner, we consider the following modification of INV KAC that stays on A_n regardless of the value of n modulo 4.

Definition 3.8. *Call the following Markov chain on A_n the two-step reversible* INV *KAC. Let X_0 be the identity permutation.*

1. *Choose $k_1 \sim \text{Unif}(\mathbb{F})$, $k_2 \sim \text{Unif}(\mathbb{F})$, $k_3 \sim \text{Unif}(\mathbb{F})$ i.i.d. Let*

$$\sigma = \text{ARK}_{k_3} \circ \text{INV} \circ \text{ARK}_{k_2} \circ \text{INV} \circ \text{ARK}_{k_1}.$$

2. *Let $X_{t+1} = \sigma \circ X_t$.*

Denote this two-step walk by $\widetilde{P}_n^{\mathrm{INV}}$. Note that $\widetilde{P}_n^{\mathrm{INV}}$ remains even as ARK_k is even for all k, and we apply INV twice. Note that, since for independent $k, \ell, m \sim \mathrm{Unif}(\mathbb{F})$, $\mathrm{ARK}_k \circ \mathrm{ARK}_\ell \sim \mathrm{ARK}_m$, it follows that each step of $\widetilde{P}_n^{\mathrm{INV}}$ corresponds to two steps of the reversible INV KAC chain, i.e. $\widetilde{P}_n^{\mathrm{INV}} = (P_n^{\mathrm{INV}})^2$.

By considering the cases $k_2 = 0$ and $k_2 \neq 0$, we may explicitly write the distribution of σ as follows:

$$\begin{aligned}\mathbb{P}[\sigma = \mathrm{ARK}_k] &= \frac{1}{n^2} && \forall k \in \mathbb{F},\\ \mathbb{P}[\sigma = \mathrm{ARK}_{k_3} \circ \mathrm{INV} \circ \mathrm{ARK}_{k_2} \circ \mathrm{INV} \circ \mathrm{ARK}_{k_1}] &= \frac{1}{n^3} && \forall k_1, k_3 \in \mathbb{F}, k_2 \in \mathbb{F}^*.\end{aligned}$$

Note that all transitions of the second type are distinct and uniquely determined by k_1, k_2, k_3. Indeed, let $a = k_2^{-1} + k_3$ and let $b = k_2^{-1} + k_1$. For all $x \neq -k_1, -k_1 - k_2^{-1}$,

$$\sigma(x) = \frac{ax + (ab - k_2^{-2})}{x + b}.$$

This clearly gives distinct functions σ, σ' for distinct selections k_1, k_2, k_3 and k_1', k_2', k_3' unless $k_2^{-1} + k_3 = k_2'^{-1} + k_3'$, $k_2^{-1} + k_1 = k_2'^{-1} + k_1'$, and $k_2 = -k_2'$; in this case, $\sigma(-k_1) = k_3 + k_2^{-1} \neq k_3 = \sigma'(-k_1)$ so σ and σ' are distinct.

Proof of Theorem 3.1 for odd characteristic. We will show in the following lemmas that

$$\mathcal{E}^{\mathrm{3cyc}}(f,f) \underset{\text{Lemma 3.10}}{\leq} 243\, \mathcal{E}^{\mathrm{3cyc,ap}}(f,f) \underset{\text{Lemma 3.12}}{\leq} 5 \cdot 243\, \mathcal{E}^{\mathrm{INV}}(f,f).$$

Let π_0 denote the uniform distribution on A_n and let π_1 denote the uniform distribution on $\mathfrak{S}_n \setminus A_n$. Note that $\widetilde{P}_n^{\mathrm{INV}} = (P_n^{\mathrm{INV}})^2$ has all nonnegative eigenvalues, so we may cite Lemma 2.11. Proceeding as in eq. (1) using Lemma 2.11 in place of Lemma 2.12,

$$\|\delta_{\mathrm{id}}(\widetilde{P}_n^{\mathrm{INV}})^t - \pi_0\|_{\mathrm{TV}} \leq \frac{1}{2}\left(|\mathfrak{S}_n| \exp\left(-\frac{t}{1215}\right)\right)^{1/2} + \frac{1}{2}\left(|\mathfrak{S}_n| \cdot \|\delta_{\mathrm{id}}(P^{\mathrm{3cyc}})^{\lfloor t/2430 \rfloor} - \pi_0\|_2^2\right)^{1/2}.$$

Using Stirling's formula for the first term and Theorem 2.8 for the second term shows that the ε-mixing time of the 2-step walk is at most $1215n(\log n + C_{2.8}\log(1/\varepsilon))$. Thus the one-step walk after t steps for even t mixes in $2430n(\log n + C_{2.8}\log(1/\varepsilon))$ steps.

Furthermore, if $n \equiv 3 \mod 4$, then since the one-step random walk is reversible with respect to the uniform distribution on A_n, it follows that for any parity of $t > 2430n(\log n + c\log(1/2\varepsilon))$, we are ε-close to the uniform distribution on A_n. If $n \equiv 1 \mod 4$, then noticing that $\pi_0 P_n^{\mathrm{INV}} = \pi_1$, we get that for odd time $2t+1$,

$$\begin{aligned}\|\delta_{\mathrm{id}}(P_n^{\mathrm{INV}})^{2t+1} - \pi_1\|_2^2 &= \|(\delta_{\mathrm{id}}(P_n^{\mathrm{INV}})^{2t} - \pi_0)P_n^{\mathrm{INV}}\|_2^2\\ &\leq \|P_n^{\mathrm{INV}}\|_{\mathrm{op}}\, \|\delta_{\mathrm{id}}(P_n^{\mathrm{INV}})^{2t} - \pi_0\|_2^2 \leq \|\delta_{\mathrm{id}}(P_n^{\mathrm{INV}})^{2t} - \pi_0\|_2^2 = \|\delta_{\mathrm{id}}(\widetilde{P}_n^{\mathrm{INV}})^{t} - \pi_0\|_2^2,\end{aligned}$$

since $\|P_n^{\mathrm{INV}}\|_{\mathrm{op}} \leq 1$.

The further improvement by a factor of 3 in characteristic $p > 3$ follows by Remark 3.11. □

We now construct a flow to analyze $\widetilde{P}_n^{\mathrm{INV}}$. We will need to consider the following variation of the random 3-cycle chain P_n^{3cyc}.

Definition 3.9 (Arithmetic progression 3-cycle Markov chain)**.** *The chain $P_n^{\mathrm{3cyc,ap}}$ on the alternating group A_n has the following transition matrix. Given the current permutation σ_t, one step in this Markov chain corresponds to drawing a uniformly random 3-cycle (i,j,k) from A_n, conditioned on the fact that i,k,j form an arithmetic progression in $\mathbb{F}_n$, that is, $k - i = j - k$. Then set*

$$\sigma_{t+1} = (i,j,k) \circ \sigma_t.$$

Our path construction shows that the underlying graph $E^{\mathrm{3cyc,ap}}$ with vertex set A_n is connected, and thus by reversibility, the unique stationary distribution of this Markov chain is uniform over A_n, i.e. $\pi_n^{\mathrm{3cyc,ap}}(x) = \frac{1}{|A_n|} = \frac{2}{n!}$. Moreover, the underlying graph is $n(n-1)$-regular. This is because there are n options for the first term of the arithmetic progression u, and $n-1$ options for the difference $-\frac{1}{v}$. Thus if $(x, y) \in E_n^{\mathrm{3cyc,ap}}$, then $P_n^{\mathrm{3cyc,ap}}(x, y) = \frac{1}{n(n-1)}$.

First, we compare the Dirichlet forms of P_n^{3cyc} and $P_n^{\mathrm{3cyc,ap}}$.

Lemma 3.10. *The Dirichlet form of $P_n^{\mathrm{3cyc,ap}}$ satisfies*

$$\mathcal{E}^{\mathrm{3cyc}}(f, f) \leq 243\, \mathcal{E}^{\mathrm{3cyc,ap}}(f, f).$$

Proof. To apply the Comparison Theorem 2.14, we will define a set of paths $\mathcal{P}_{xy}$ for each edge $(x, y) \in E^{\mathrm{3cyc}}$ and assign flow $F(\cdot)$ to each path.

Our construction of paths in this case is quite simple. We will assign exactly one path to each such edge (x, y). In particular, let $y = (i, k, j) \circ x$ for some triple of pairwise distinct elements i, j, k. Then $\mathcal{P}_{xy} = \{\gamma_{xy}\}$, and this path is chosen according to the following two cases:

1. The elements i, j, k form an arithmetic progression. This means that either $k - i = j - k$, or $i - j = k - i$, or $j - k = i - j$. Then (i, k, j) is also in $E_n^{\mathrm{3cyc,ap}}$ and we set

$$\gamma_{xy} = (x, (i, k, j) \circ x = y).$$

   The flow is

$$F(\gamma_{xy}) = P_n^{\mathrm{3cyc}}(x, y) \cdot \pi_n^{\mathrm{3cyc}}(x, y) = \frac{1}{2\binom{n}{3}} \cdot \frac{2}{n!} = \frac{6}{n(n-1)(n-2) \cdot n!}.$$

2. The elements i, j, k do not form an arithmetic progression. Let $u = \frac{i+j}{2}, v = \frac{j+k}{2}, w = \frac{k+i}{2}$. Then $E_n^{\mathrm{3cyc,ap}}$ contains the distinct 3-cycles

$$C_1 = (i, j, u), \quad C_2 = (j, k, v), \quad C_3 = (k, i, w).$$

   We assign to (x, y) the length-9 path

$$\begin{aligned}\gamma_{xy} = (x, &\underbrace{C_1 \circ x}_{a_1}, \underbrace{C_3 \circ a_1}_{a_2}, \underbrace{C_3 \circ a_2}_{a_3}, \underbrace{C_2 \circ a_3}_{a_4}, \\ &\underbrace{C_2 \circ a_4}_{a_5}, \underbrace{C_3 \circ a_5}_{a_6}, \underbrace{C_1 \circ a_6}_{a_7}, \underbrace{C_2 \circ a_7}_{a_8}, C_2 \circ a_8 = y).\end{aligned}$$

   The proof that this path connects x to y is deferred to Lemma C.2. By relabeling i, j, k, we can see

$$C_2 \circ C_1 \circ C_1 \circ C_3 \circ C_3 \circ C_1 \circ C_2 \circ C_3 \circ C_3 = (j, k, i),$$

   which is the same permutation as (i, j, k). We may rotate the indices one further to get (k, i, j), producing three distinct paths. We will distribute the flow evenly among these paths, assigning to each $F(\gamma_{xy}) = 2/(n(n-1)(n-2) \cdot n!)$.

The comparison constant we get is thus

$$
\begin{aligned}
A(F) &= \max_{(a,b)\in E^{3\mathrm{cyc,ap}}}\left\{\frac{1}{\pi^{3\mathrm{cyc,ap}}(a)\cdot P^{3\mathrm{cyc,ap}}(a,b)}\sum_{\gamma\in\mathcal{P}(a,b)}|\gamma|\cdot F(\gamma)\right\}\\
&= \max_{(a,b)\in E^{3\mathrm{cyc,ap}}}\left\{\frac{n!}{2}\cdot n(n-1)\cdot\left(1\cdot\frac{6}{n(n-1)(n-2)n!}+81(n-3)\cdot 3\cdot\frac{2}{n(n-1)(n-2)n!}\right)\right\}\\
&= \frac{3}{n-2}+243\cdot\frac{n-3}{n-2}\le 243.
\end{aligned}
$$

We used the fact that the number of paths γ in $\mathcal{P}(a,b)$ is at most $81(n-3)$. This is because $\mathcal{P}(a,b)$ contains paths of length 1 and 9. The set $\mathcal{P}(a,b)$ contains at most 1 path of length 1 for any $(a,b)\in E^{3\mathrm{cyc,ap}}$.

Moreover, length-9 paths γ in $\mathcal{P}(a,b)$ must be using the edge (a,b) as their cycle C_1, C_2, or C_3 and at some step $s\in\{1,2,\ldots,9\}$. Let $(p,q,r)=ba^{-1}$ be the step taken. Without loss of generality, suppose the cycle is C_1; then we know that the ordered pair $(i,j)\in\{(p,q),(q,r),(r,p)\}$, giving three choices for (i,j), and so $3(n-2)$ choices (i,j,k) (as $k\neq p,q,r$). If the cycle is C_2 or C_3, then we just reveal a different two of i,j,k first and still have $3(n-2)$ choices. Note also that knowing which C_ℓ is at which time s informs us which of the three kinds of paths we constructed we are currently on. Thus there are at most $81(n-3)$ total paths in $\mathcal{P}(a,b)$. □

Remark 3.11 (Characteristic greater than 3)**.** *Given an unordered 3-term arithmetic progression* $\{z,z',z''\}$*, we may recover the middle element by noting that it is the arithmetic mean* $\frac{1}{3}(z+z'+z'')$ *if* $3\neq 0$*. Thus we know which of* $\{(p,q),(q,r),(r,p)\}$ *must be* (i,j)*, improving the above by a factor of 3.*

Finally, we compare the Dirichlet forms of $P_n^{3\mathrm{cyc,ap}}$ and $\widetilde{P}_n^{\mathrm{INV}}$.

Lemma 3.12. *The Dirichlet form of* $\widetilde{P}_n^{\mathrm{INV}}$ *satisfies*

$$\mathcal{E}^{3\mathrm{cyc,ap}}(f,f)\le 5\,\mathcal{E}^{\widetilde{\mathrm{INV}}}(f,f)$$

Proof. We construct a $(\widetilde{P}_n^{\mathrm{INV}}, P_n^{3\mathrm{cyc,ap}})$-flow F with $A(F)\le 5$.

Observe that for any $u\in\mathbb{F}$ and $v\in\mathbb{F}^*$, the sequence

$$\psi_{\mathrm{odd}}(u,v)=[[u,v,-2v^{-1},v,-u-2v^{-1}]] \tag{2}$$

results in the 3-cycle $(-u,-u-2v^{-1},-u-v^{-1})$, which is a step of $P_n^{3\mathrm{cyc,ap}}$. We prove this in Appendix C.

Fix any $u\in\mathbb{F}$, $v\in\mathbb{F}^*$. For any $k\in\mathbb{F}$,

$$[[-u,-v,-k]],[[-2v^{-1}+k,v,-u-2v^{-1}]]$$

composes to give the same arithmetic progression 3-cycle $(-u,-u-2v^{-1},-u-v^{-1})$. This provides a representation of this arithmetic progression 3-cycle using two steps of $\widetilde{P}_n^{\mathrm{INV}}$. Note that since $v\neq 0$, neither of the two steps used are translations.

Define the flow F by averaging over all of these length two paths (there are n such paths, one for each choice of k). In particular, for each such path γ, let

$$F(\gamma)=\frac{1}{n}P_n^{3\mathrm{cyc,ap}}(x,y)\pi(x)=\frac{1}{n^2(n-1)n!}.$$

The comparison constant is then

$$A(F)=\max_{(a,b)\in E^{\mathrm{tar}}}\left\{\frac{1}{\pi^{\mathrm{tar}}(a)P^{\mathrm{tar}}(a,b)}\sum_{\gamma\in\mathcal{P}(a,b)}|\gamma|\cdot F(\gamma)\right\}=\left(2+\frac{2}{n-1}\right)\max_{(a,b)\in E^{\mathrm{tar}}}|\mathcal{P}(a,b)|.$$

We have used that $P^{\mathrm{tar}}(a,b) = 1/n^3$, which is valid since the edges used in our paths are not translations.

We will show that $|\mathcal{P}(a,b)| = 2$ whenever $t := ba^{-1}$ is not a translation. In this case, there exist $x, y, z \in \mathbb{F}$ with $y \neq 0$ such that

$$t = \mathrm{ARK}_x \circ \mathrm{INV} \circ \mathrm{ARK}_y \circ \mathrm{INV} \circ \mathrm{ARK}_z\,.$$

By construction, all paths $\gamma \in \mathcal{P}(a,b)$ have length 2. If (a,b) is the first transition in γ, then $v = y$ and $u = z$, and so the transition in $P_n^{\mathrm{3cyc,ap}}$ is exactly $(-z, -z-2y^{-1}, -z-y^{-1})$. If (a,b) is the second transition, then $v = y$ and $u = -x-2y^{-1}$ and so the transition in $P_n^{\mathrm{3cyc,ap}}$ is exactly $(x+2y^{-1}, x, x+y^{-1})$. Thus $|\mathcal{P}(a,b)| = 2$ and so $A(F) = 4(1+1/(n-1)) \leq 5$.

The conclusion now follows by Theorem 2.14. $\square$

4 Lower bound

In this section, we prove tight lower bounds on the number of rounds it takes INV KAC to achieve 4-wise independence.

First, we show that $\Omega(n\log(1/\varepsilon))$ rounds are required for INV KAC to become ε-approximately 4-wise independent (in the sense of Definition 2.1). The precise statement is as follows.

Theorem 4.1. *Let $\varepsilon = \varepsilon(n) \in (0,1)$ be such that $\varepsilon n \to \infty$ as $n \to \infty$. If $t < \frac{1}{5}n\log(1/\varepsilon)$, then X_t is not $(1+o_\varepsilon(1))\varepsilon$-approximately 4-wise independent. (Here, the $o_\varepsilon(1)$ term goes to 0 as $\varepsilon \to 0$.)*

Combining this with our upper bound, we see that in order to be ε-approximately 4-wise independent (for $\varepsilon n \to \infty$), the INV KAC walk requires τ steps where

$$\Theta(n\log(1/\varepsilon)) \leq \tau \leq \Theta(n(\log(1/\varepsilon) + \log(n))).$$

In particular, this is optimal up to a constant if $\varepsilon = n^{-\alpha}$ for $0 < \alpha < 1$.

Finally, we show that for any 4-wise independent permutation σ, INV KAC needs at least $\Omega(n\log n)$ rounds to get within $(1 - o_n(1))$ in total variation distance of σ. Taking σ to be the uniform distribution on permutations of a fixed sign provides a matching lower bound to our mixing time upper bound (Theorem 3.1).

Theorem 4.2. *Let σ be a 4-wise independent permutation. For any $t < \frac{1}{7}n\log n$, the total variation distance between the* INV *KAC walk X_t and σ is $(1 - o_n(1))$.*

4.1 Möbius transformation approximation

Define the set of *modified Möbius transformations*:

$$\mathcal{M}_n := \left\{ m(x) = (ax+b)(cx+d)^{n-2} \,\middle|\, a,b,c,d \in \mathbb{F},\ ad-bc \neq 0 \right\}.$$

Note that $\mathcal{M}_n$ consists of functions $m : \mathbb{F} \to \mathbb{F}$ which are generally not permutations. Our proofs rely on two lemmas. First, we show that for $t < \frac{1}{5}n\log(1/\varepsilon)$, X_t is approximated too well (compared to a 4-wise independent permutation, see Lemma 4.4) by some element of $\mathcal{M}_n$.

Lemma 4.3. *Let $\varepsilon = \varepsilon(n) \in (0,1)$ be such that $\varepsilon^{1/4}n \to \infty$ as $n \to \infty$. Then for $t < \frac{1}{5}n\log(1/\varepsilon)$, there exists $m_t \in \mathcal{M}_n$ such that*

$$d_{\mathrm{Hamming}}(m_t, X_t) \leq n - \varepsilon^{1/4}n$$

except with probability $O(1/\log(1/\varepsilon))$.

Proof. Let $k_1, \dots, k_t \in \mathbb{F}$ and $g_i(x) = (x - k_i)^{n-2}$ be the functions chosen in step (2) of the algorithm so that $X_t = g_1 \circ \cdots \circ g_t$. Let $X_i = g_1 \circ \cdots \circ g_i$, with X_0 equal to the identity function. We will iteratively create a sequence of elements $Y_0, \dots, Y_t \in \mathcal{M}_n$ and a sequence of sets $B_1, \dots, B_t \subset \mathbb{F}$ such that Y_i agrees with X_i outside of B_i.

We begin by taking Y_0 to be the identity map (i.e. $a = d = 1$ and $b = c = 0$) and $B_0 = \emptyset$, for which the claim is true by definition of X_0. For $i = 1, \dots, t$, given $Y_{i-1}(x) = (ax + b)(cx + d)^{n-2}$, let

$$Y_i(x) = (bx + (a - bk_i))(dx + (c - dk_i))^{n-2}.$$

Note that the determinant of Y_i is

$$b(c - dk_i) - (a - bk_i)d = -(ad - bc) \neq 0,$$

so $Y_i \in \mathcal{M}_n$. Moreover, if $x \neq k_i$, then it follows from Fermat's little theorem that

$$\begin{aligned}
(Y_{i-1} \circ g_i)(x) &= (a(x - k_i)^{n-2} + b)(c(x - k_i)^{n-2} + d)^{n-2} \\
&= (x - k_i)^{n-2}(a + b(x - k_i))\big((x - k_i)^{n-2}(c + d(x - k_i))\big)^{n-2} \\
&= (x - k_i)^{(n-2)+(n-2)^2}(bx + (a - bk_i))(dx + (c - dk_i))^{n-2} \\
&= (bx + (a - bk_i))(dx + (c - dk_i))^{n-2} = Y_i(x).
\end{aligned}$$

Let $B_i = g_i^{-1}(B_{i-1}) \cup \{k_i\}$. Since Y_{i-1} agrees with X_{i-1} outside of B_{i-1}, Y_i agrees with $Y_{i-1} \circ g_i$ outside $\{k_i\}$, and since $X_i = X_{i-1} \circ g_i$, it follows that X_i agrees with Y_i outside $g_i^{-1}(B_{i-1}) \cup \{k_i\} = B_i$.

Thus, to complete the proof, it suffices to show that with probability $1 - O(\log(1/\varepsilon))$ over the choice of $k_1, \dots, k_t$ drawn independently and uniformly at random from $\mathbb{F}$, $|B_t| \leq n - \varepsilon^{1/4}n$. This follows by a standard coupon-collector argument.

Formally, define the auxiliary sequence of sets $C_i = (g_1 \circ g_2 \circ \cdots \circ g_i)(B_i)$. Recalling the recurrence for B_i gives the recurrence

$$C_i = (g_1 \circ g_2 \circ \cdots \circ g_i)(g_i^{-1}(B_{i-1}) \cup \{k_i\}) = C_{i-1} \cup \{(g_1 \circ g_2 \circ \cdots \circ g_{i-1})(0)\},$$

since $g_i(k_i) = 0$. Thus C_i is a chain of sets. Note that $C_0 = \emptyset$, $C_1 = \{0\}$, and it is clear inductively that C_i only depends on $k_1, \dots, k_{i-1}$. We produce C_i by adding to C_{i-1} the element $(g_1 \circ g_2 \circ \cdots \circ g_{i-1})(0)$, which (for $i > 1$) is uniformly distributed over $\mathbb{F}$ (conditioned on $k_1, \dots, k_{i-2}$) since k_{i-1} is uniform over $\mathbb{F}$ and independent of $k_1, \dots, k_{i-2}$.

Now, C_i is a standard coupon-collector process and $|C_i| = |B_i|$. Let $\tau_i = \min\{s : |C_s| \geq i\} - \min\{s : |C_s| \geq i - 1\}$. Then $\tau_i \sim \mathrm{Geom}((n + 1 - i)/n)$ are independent, so we may compute

$$\begin{aligned}
\mathbb{E}\left[\sum_{i=1}^{n-\varepsilon^{1/4}n} \tau_i\right] &= \sum_{i=1}^{n-\varepsilon^{1/4}n} \frac{n}{n+1-i} = n\left(\sum_{j=1}^{n} \frac{1}{j} - \sum_{j=1}^{\varepsilon^{1/4}n} \frac{1}{j}\right) = n\left(\frac{1}{4}\log(1/\varepsilon) + O(1)\right), \\
\mathrm{Var}\left(\sum_{i=1}^{n-\varepsilon^{1/4}n} \tau_i\right) &\leq \sum_{i=1}^{n-\varepsilon^{1/4}n} \left(\frac{n}{n+1-i}\right)^2 \leq n^2 \sum_{j=1}^{\infty} \frac{1}{j^2} = \frac{\pi^2}{6} n^2.
\end{aligned}$$

The first line uses the assumption $\varepsilon^{1/4}n \to \infty$. Therefore, by Chebyshev's inequality,

$$\mathbb{P}[|B_t| > n - \varepsilon^{1/4}n] \leq \mathbb{P}\left[\sum_{i=1}^{n-\varepsilon^{1/4}n} \tau_i \leq \frac{1}{5} n \log(1/\varepsilon)\right] \leq \frac{400 \cdot \pi^2}{6\log(1/\varepsilon)},$$

as desired. □

Next, we show that for any 4-wise independent permutation σ, the probability that the image on 4 distinct points $T = \{x_1, x_2, x_3, x_4\}$ fits a modified Möbius transformation is $O(1/n)$. Formally, for $T \in \binom{\mathbb{F}}{4}$, let B_T denote the event (depending on σ) that there exists $m \in \mathcal{M}_n$ with $\sigma \restriction_T = m \restriction_T$.

Lemma 4.4. *With notation as above,*

$$\mathbb{P}[B_T] = O(n^{-1}).$$

Proof. Note that the standard Möbius transformation

$$\begin{aligned} m : \mathbb{F} \cup \{\infty\} &\to \mathbb{F} \cup \{\infty\}, \\ x &\mapsto \frac{ax+b}{cx+d} \end{aligned}$$

is a permutation, and is completely determined by any 3 points. The modified transformation $(ax+b)(cx+d)^{n-2}$ on $\mathbb{F}$ agrees everywhere except $-d/c \mapsto 0$ and a/c is not in the range. By the law of total probability

$$\begin{aligned} \mathbb{P}[B_T] &= \mathbb{P}[B_T \mid 0 \notin \sigma(T)]\mathbb{P}[0 \notin \sigma(T)] + \mathbb{P}[B_T \mid 0 \in \sigma(T)]\mathbb{P}[0 \in \sigma(T)] \\ &\le \mathbb{P}[B_T \mid 0 \notin \sigma(T)] + \mathbb{P}[0 \in \sigma(T)]. \end{aligned}$$

By a union bound and using 4-wise independence, $\mathbb{P}[0 \in \sigma(T)] \le 8/n$. If $0 \notin \sigma(T)$, then let $T = \{x_1, x_2, x_3, x_4\}$ and reveal $\sigma(x_1), \sigma(x_2), \sigma(x_3)$. There is exactly one Möbius transformation defined by these three points $(x_i, \sigma(x_i))$ and so one modified Möbius transformation m^* since $0 \notin \sigma(x_i)$. Thus we must bound $\mathbb{P}[\sigma(x_4) = m^*(x_4)]$. By 4-wise independence, $\sigma(x_4) \sim \mathrm{Unif}(\mathbb{F} \setminus \{0, \sigma(x_1), \sigma(x_2), \sigma(x_3)\})$ after conditioning on these values, so $\mathbb{P}[\sigma(x_4) = m^*(x_4)] = 1/(n-4)$. Thus $\mathbb{P}[B_T] \le 10/n$ for $n \ge 8$. ☐

4.2 Derivation of lower bounds

Given the preceding lemmas, the proof of Theorem 4.1 is essentially immediate.

Proof of Theorem 4.1. For a 4-tuple of distinct elements $T = \{a_1, a_2, a_3, a_4\} \subseteq \mathbb{F}$ and a permutation π, let $A_T(\pi) = 1$ if there exists a modified Möbius transformation $m \in \mathcal{M}_n$ such that $m(a_i) = \pi(a_i)$ for all $i \in [4]$, and let $A_T(\pi) = 0$ otherwise.

We claim that there exists a 4-tuple T^* such that if $t < \frac{1}{5} n \log(1/\varepsilon)$, then

$$\mathbb{E}[A_{T^*}(X_t)] \ge \varepsilon\bigg(1 + O\bigg(\frac{1}{\log(1/\varepsilon)} + \frac{1}{\varepsilon^{1/4}n} + \frac{1}{n}\bigg)\bigg)$$

Indeed, we have

$$\begin{aligned} \mathbb{E}_{T \sim \binom{\mathbb{F}}{4}, X_t}[A_T(X_t)] &\ge \mathbb{E}_{T, X_t}[A_T(X_t) \mid \text{Lemma 4.3 holds}] \cdot \mathbb{P}[\text{Lemma 4.3 holds}] \\ &\ge \frac{\binom{\varepsilon^{1/4}n}{4}}{\binom{n}{4}} \cdot \bigg(1 - O\bigg(\frac{1}{\log(1/\varepsilon)}\bigg)\bigg) = \varepsilon\bigg(1 + O\bigg(\frac{1}{\log(1/\varepsilon)} + \frac{1}{\varepsilon^{1/4}n} + \frac{1}{n}\bigg)\bigg). \end{aligned}$$

Therefore, the probabilistic method implies the existence of a T^* satisfying the desired bound. On the other hand, by Lemma 4.4, for a 4-wise independent permutation σ, $\mathbb{P}[A_{T^*}(\sigma)] = O(1/n)$. Therefore,

$$\begin{aligned} \|X_t \restriction_{T^*} - \sigma \restriction_{T^*}\|_{\mathrm{TV}} &\ge \mathbb{E}[A_{T^*}(X_t)] - \mathbb{E}[A_{T^*}(\sigma)] \\ &\ge \varepsilon\bigg(1 + O\bigg(\frac{1}{\log(1/\varepsilon)} + \frac{1}{\varepsilon^{1/4}n}\bigg)\bigg) - O\bigg(\frac{1}{n}\bigg) \\ &= \bigg(1 + O\bigg(\frac{1}{\log(1/\varepsilon)} + \frac{1}{\varepsilon n}\bigg)\bigg)\varepsilon, \end{aligned}$$

which is $(1 + o_\varepsilon(1))\varepsilon$ for $\varepsilon n \to \infty$. ☐

Proof of Theorem 4.2. We construct a test $\mathcal{D}$ which, with high probability, can distinguish between X_t for $t < n\log n/7$ and a 4-wise independent permutation σ. On input a permutation π:

1. Compute
$$Z_\pi = \#\left\{\{x_1, x_2, x_3, x_4\} \in \binom{\mathbb{F}}{4} : \exists m \in \mathcal{M}_n \text{ such that } m(x_i) = \pi(x_i) \text{ for all } i \in [4]\right\}.$$
2. If $Z_\pi \geq \binom{n^{0.8}}{4}$, ACCEPT.
3. Else, REJECT.

We claim that for $t < \frac{1}{7}n\log n$:
$$\mathbb{P}[\mathcal{D}(X_t) = \text{ACCEPT}] \geq 1 - o_n(1).$$
Indeed, by Lemma 4.3, with probability $1 - o_n(1)$, X_t agrees with some $m_t \in \mathcal{M}_n$ on $\geq n^{0.8}$ points. Whenever this holds, we have
$$Z_{X_t} \geq \#\left\{\{x_1, x_2, x_3, x_4\} \in \binom{\mathbb{F}}{4} : m_t(x_i) = X_t(x_i) \text{ for all } i \in [4]\right\} \geq \binom{n^{0.8}}{4},$$
so that X_t is accepted by $\mathcal{D}$.

On the other hand, for a 4-wise independent permutation σ, it follows by Lemma 4.4, Markov's inequality and linearity of expectation that
$$\mathbb{P}\left[Z_\sigma \geq \binom{n^{0.8}}{4}\right] \leq \frac{\mathbb{E}[Z_\sigma]}{\binom{n^{0.8}}{4}} = \left(\binom{n}{4} \cdot \frac{10}{n}\right) \cdot \frac{24}{(1+o(1))n^{3.2}} = O(n^{-0.2}).$$
Therefore, we have
$$\|X_t - \sigma\|_{\mathrm{TV}} \geq \mathbb{P}[\mathcal{D}(X_t) = \text{ACCEPT}] - \mathbb{P}[\mathcal{D}(\sigma) = \text{ACCEPT}] \geq 1 - o_n(1). \qquad \square$$

Acknowledgments

The authors would like to thank Alistair Sinclair and Lucas Gretta for very helpful and insightful discussions regarding the functional analysis of Markov chains. VJ's research was supported by NSF CAREER award DMS-2237646. CM's research was supported by NSF award ECCS-2217023 through the IDEAL institute. AP's research was supported by DARPA under Agreement No. HR00112020023. ST's research was supported by NSF grants CNS-2026774, CNS-2154174, CNS-2426905, and a Stellar Development Foundation Academic Research Award. VV's research was supported in part by DARPA under Agreement Number HR00112020023, NSF CNS-2154149, a Simons Investigator award and a Ford Foundation Chair at MIT.

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A Comparison for alternating walks: Proof of Lemma 2.12

We prove Lemma 2.12. This follows very similarly to the proof of [DSC93a, Lemma 5]. The only insight is that bipartite graphs have symmetric spectra, so we need not worry about large negative eigenvalues besides the unique eigenvalue -1.

Proof of Lemma 2.12. Let $p = p^{\text{tar}}$ and $\widetilde{p} = p^{\text{ref}}$ be the probability measures generating the two walks. Let $p^{(*t)}$ denote the t-fold convolution and let $1 = \lambda_0 \geq \lambda_1 \geq \cdots \geq \lambda_{|G|-1} = -1$ denote the eigenvalues of the corresponding operator.

Let $\operatorname{sgn} : G \to \{\pm 1\}$ denote the group homomorphism with kernel H. Then

$$\begin{aligned}
\|p^{(*t)} - \pi_t\|_2^2 &= \sum_{g\in G}\left(p^{(*t)}(g) - \frac{1+(-1)^t\operatorname{sgn}(g)}{|G|}\right)^2 \\
&= \frac{2}{|G|} + \sum_{g\in G} p^{(*t)}(g)^2 - 2\sum_{g\in G} p^{(*t)}(g)\frac{1+\operatorname{sgn}(g)(-1)^t}{|G|} \\
&= \sum_{g\in G} p^{(*t)}(g)^2 - \frac{2\cdot(-1)^t}{|G|}\Big[p^{(*t)}(H) - p^{(*t)}(G\setminus H)\Big].
\end{aligned}$$

Notice that by periodicity, $p^{(*t)}(H) - p^{(*t)}(G\setminus H) = (-1)^t$. Thus we get

$$\begin{aligned}
\|p^{(*t)} - \pi_t\|_2^2 &= -\frac{2}{|G|} + \sum_{g\in G} p^{(*t)}(g)^2 \\
&= -\frac{2}{|G|} + \frac{1}{|G|}\sum_{i=0}^{|G|-1}\lambda_i^{2t} \\
&= \frac{1}{|G|}\sum_{i=1}^{|G|-2}\lambda_i^{2t} \\
&= \frac{2}{|G|}\sum_{i=1}^{|G|/2-1}\lambda_i^{2t},
\end{aligned}$$

where the last equality uses that bipartite graphs have symmetric spectra. Now, using the inequality $1-x \leq e^{-x}$, [DSC93a, Lemma 4], $1-x \geq e^{-2x}$ for $0 \leq x \leq 1/2$, and proceeding as in the proof of [DSC93a, Lemma 5], we have

$$\begin{aligned}
\|p^{(*t)} - \pi_t\|_2^2 &\le \frac{2}{|G|} \sum_{i=1}^{|G|/2-1} e^{-2t(1-\lambda_i)} \\
&\le \frac{2}{|G|} \sum_{i=1}^{|G|/2-1} e^{-2t/A(1-\widetilde{\lambda}_i)} \\
&\le e^{-t/A} + \frac{2}{|G|} \sum_{i=1}^{|G|/2-1} (1-(1-\widetilde{\lambda}_i))^{t/2A} \\
&\le e^{-t/A} + \|\widetilde{p}^{(*\lfloor t/2A \rfloor)} - \pi_{\lfloor t/2A \rfloor}\|_2^2.
\end{aligned}$$

□

B Proof of Lemma 3.6

In this section, we prove Lemma 3.6. In fact, we will prove the following more general statement, which doesn't require the characteristic of $\mathbb{F}_n$ to be 2, but instead only requires that existence of a square root of -1. For the rest of this section, fix $\iota \in \mathbb{F} := \mathbb{F}_n$ to be an element with $\iota^2 = -1$.

Lemma B.1 (Generalization of Lemma 3.6)**.** *For any $u, v, w \in \mathbb{F}_n$ such that $uv \neq -\iota$, $uv + u \neq -\iota$, and $u \neq 0$, we can generate the transposition $\left(0, \frac{-\iota}{(uv+\iota)(uv+u+\iota)}\right)$ using the following sequence $\psi_{\text{even}}(u, v, w)$ of* ARK *and* INV *operations*

$$\psi_{\text{even}}(u,v,w) = \left[\left[2u + \frac{v}{uv+\iota},\quad u,\quad -\iota v,\quad \mathrm{INV}(w)+\iota,\quad -w,\right.\right.$$
$$\left.\left.\mathrm{INV}(w),\quad -w,\quad -\mathrm{INV}(w),\quad -w+\iota,\quad \iota v+\iota,\quad -u,\quad \frac{v}{uv+\iota}\right]\right].$$

The proof of Lemma B.1 relies on two intermediate lemmas, Lemma B.2 and Lemma B.3.

Lemma B.2. *Suppose $u, v \in \mathbb{F}_n$ such that $u \neq 0$, $uv \neq -\iota$, and $u(v+1) \neq -\iota$. Then the sequence $\gamma(u, v)$ of* AddRoundKey *and* INV *S-box operations*

$$\gamma(u,v) = [[u, u, -\iota v, \iota, \iota, \iota v + \iota, -u, -u]]$$

gives the transposition

$$\left(-u - \frac{v}{uv+\iota}, -u - \frac{v+1}{uv+u+\iota}\right).$$

Proof. We break this into three cases.

Case 1 (general) Suppose $x \neq -u, -u-1/u, -u-v/(uv+\iota), -u-(v+1)/(uv+u+\iota)$. Then

$$\begin{aligned}
x &\xrightarrow{\mathrm{ARK}_u} x+u \xrightarrow{\mathrm{INV}} \frac{1}{x+u} \xrightarrow{\mathrm{ARK}_u} \frac{ux+u^2+1}{x+u} \xrightarrow{\mathrm{INV}} \frac{x+u}{ux+u^2+1} \\
&\xrightarrow{\mathrm{ARK}_{-\iota v}} \frac{(-\iota uv+1)x+u^2v-\iota v+u}{ux+u^2+1} \xrightarrow{\mathrm{INV}} \frac{ux+u^2+1}{(-\iota uv+1)x-\iota u^2v-\iota v+u} \\
&\xrightarrow{\mathrm{ARK}_\iota} \frac{(uv+\iota+u)x+u^2+1+u^2v+v+\iota u}{(-\iota uv+1)x-\iota u^2v-\iota v+u} \xrightarrow{\mathrm{INV}} \frac{(-\iota uv+1)x-\iota u^2v-\iota v+u}{(uv+\iota+u)x+u^2+1+u^2v+v+\iota u} \\
&\xrightarrow{\mathrm{ARK}_\iota} \frac{\iota ux+\iota u^2+\iota}{(uv+\iota+u)x+u^2+1+u^2v+v+\iota u} \xrightarrow{\mathrm{INV}} \frac{(uv+\iota+u)x+u^2+1+u^2v+v+\iota u}{\iota ux+\iota u^2+\iota} \\
&\xrightarrow{\mathrm{ARK}_{\iota v+\iota}} \frac{\iota x+\iota u}{\iota ux+\iota u^2+\iota} \xrightarrow{\mathrm{INV}} \frac{ux+u^2+1}{x+u} \xrightarrow{\mathrm{ARK}_{-u}} \frac{1}{x+u} \xrightarrow{\mathrm{INV}} x+u \xrightarrow{\mathrm{ARK}_{-u}} x.
\end{aligned}$$

Case 2 (other fixed points) For $x=-u$,

$$\begin{aligned}
&-u \xrightarrow{\mathrm{ARK}_u} 0 \xrightarrow{\mathrm{INV}} 0 \xrightarrow{\mathrm{ARK}_u} u \xrightarrow{\mathrm{INV}} \frac{1}{u} \xrightarrow{\mathrm{ARK}_{-\iota v}} \frac{-\iota uv+1}{u} \xrightarrow{\mathrm{INV}} \frac{u}{-\iota uv+1} \xrightarrow{\mathrm{ARK}_\iota} \frac{u+uv+\iota}{-\iota uv+1} \xrightarrow{\mathrm{INV}} \\
&\frac{-\iota uv+1}{u+uv+\iota} \xrightarrow{\mathrm{ARK}_\iota} \frac{\iota u}{u+uv+\iota} \xrightarrow{\mathrm{INV}} \frac{u+uv+\iota}{\iota u} \xrightarrow{\mathrm{ARK}_{\iota v+\iota}} \frac{\iota}{\iota u} \xrightarrow{\mathrm{INV}} u \xrightarrow{\mathrm{ARK}_{-u}} 0 \xrightarrow{\mathrm{INV}} 0 \xrightarrow{\mathrm{ARK}_{-u}} -u.
\end{aligned}$$

Similarly, for $x=-u-1/u$,

$$\begin{aligned}
&-u-\frac{1}{u} \xrightarrow{\mathrm{ARK}_u} -\frac{1}{u} \xrightarrow{\mathrm{INV}} -u \xrightarrow{\mathrm{ARK}_u} 0 \xrightarrow{\mathrm{INV}} 0 \xrightarrow{\mathrm{ARK}_{-\iota v}} -\iota v \xrightarrow{\mathrm{INV}} \frac{1}{-\iota v} \xrightarrow{\mathrm{ARK}_\iota} \frac{v+1}{-\iota v} \xrightarrow{\mathrm{INV}} \frac{-\iota v}{v+1} \\
&\xrightarrow{\mathrm{ARK}_\iota} \frac{\iota}{v+1} \xrightarrow{\mathrm{INV}} \frac{v+1}{\iota} \xrightarrow{\mathrm{ARK}_{\iota v+\iota}} 0 \xrightarrow{\mathrm{INV}} 0 \xrightarrow{\mathrm{ARK}_{-u}} -u \xrightarrow{\mathrm{INV}} -\frac{1}{u} \xrightarrow{\mathrm{ARK}_{-u}} -u-\frac{1}{u}.
\end{aligned}$$

Case 3 (transposition) However, for $x=-u-v/(uv+\iota)$,

$$\begin{aligned}
&-u-\frac{v}{uv+\iota} \xrightarrow{\mathrm{ARK}_u} \frac{-v}{uv+\iota} \xrightarrow{\mathrm{INV}} \frac{uv+\iota}{-v} \xrightarrow{\mathrm{ARK}_u} \frac{1}{\iota v} \xrightarrow{\mathrm{INV}} \iota v \xrightarrow{\mathrm{ARK}_{-\iota v}} 0 \xrightarrow{\mathrm{INV}} 0 \xrightarrow{\mathrm{ARK}_\iota} \iota \\
&\xrightarrow{\mathrm{INV}} -\iota \xrightarrow{\mathrm{ARK}_\iota} 0 \xrightarrow{\mathrm{INV}} 0 \xrightarrow{\mathrm{ARK}_{\iota v+\iota}} \iota v+\iota \xrightarrow{\mathrm{INV}} \frac{-\iota}{v+1} \xrightarrow{\mathrm{ARK}_{-u}} \frac{-uv-u-\iota}{v+1} \\
&\xrightarrow{\mathrm{INV}} \frac{v+1}{-uv-u-\iota} \xrightarrow{\mathrm{ARK}_{-u}} -u-\frac{v+1}{uv+u+\iota}.
\end{aligned}$$

As this is a permutation and all other points are fixed, it exchanges these two points. Thus we have the desired transposition in this case as well. □

Lemma B.3. *The following two sequences result in the same permutation.*

$$[[\iota,\iota]] = [[\mathrm{INV}(w)+\iota, -w, \mathrm{INV}(w), -w, -\mathrm{INV}(w), -w+\iota]].$$

Proof. If $w=0$ then this is obviously true, as the middle four steps collapse to a single INV operation. We will assume $w \neq 0$ for the remainder of the proof.

Observe for $x \neq -\iota$

$$x \xrightarrow{\mathrm{ARK}_\iota} x+\iota \xrightarrow{\mathrm{INV}} \frac{1}{x+\iota} \xrightarrow{\mathrm{ARK}_\iota} \frac{\iota x}{x+\iota}$$

and for $x \neq -\iota$, $x \neq -\iota - 1/w$,

$$x \xrightarrow{\text{ARK}_{1/w+\iota}} \frac{wx+\iota w+1}{w} \xrightarrow{\text{INV}} \frac{w}{wx+\iota w+1} \xrightarrow{\text{ARK}_{-w}} \frac{-w^2x-\iota w^2}{wx+\iota w+1} \xrightarrow{\text{INV}} \frac{wx+\iota w+1}{-w^2x-\iota w^2}$$
$$\xrightarrow{\text{ARK}_{1/w}} \frac{1}{-w^2x-\iota w^2} \xrightarrow{\text{INV}} -w^2x-\iota w^2 \xrightarrow{\text{ARK}_{-w}} -w^2x-\iota w^2-w \xrightarrow{\text{INV}} \frac{1}{-w^2x-\iota w^2-w}$$
$$\xrightarrow{\text{ARK}_{1/w}} \frac{-x-\iota}{-wx-\iota w-1} \xrightarrow{\text{INV}} \frac{wx+\iota w+1}{x+\iota} \xrightarrow{\text{ARK}_{-w+\iota}} \frac{\iota x}{x+\iota}.$$

We now evaluate both for $x = -\iota$. For the first,

$$-\iota \xrightarrow{\text{ARK}_{\iota}} 0 \xrightarrow{\text{INV}} 0 \xrightarrow{\text{ARK}_{\iota}} \iota$$

and the second

$$-\iota \xrightarrow{\text{ARK}_{1/w+\iota}} \frac{1}{w} \xrightarrow{\text{INV}} w \xrightarrow{\text{ARK}_{-w}} 0 \xrightarrow{\text{INV}} 0 \xrightarrow{\text{ARK}_{1/w}} \frac{1}{w} \xrightarrow{\text{INV}} w \xrightarrow{\text{ARK}_{-w}} 0 \xrightarrow{\text{INV}} 0$$
$$\xrightarrow{\text{ARK}_{1/w}} \frac{1}{w} \xrightarrow{\text{INV}} w \xrightarrow{\text{ARK}_{-w+\iota}} \iota.$$

As both are permutations and they now agree on $n-1$ points, they must agree on the last. □

With these intermediate results in hand, we can now prove Lemma B.1.

Proof of Lemma B.1. From Lemmas B.2 and B.3 we know that the sequence

$$\psi'_{\text{even}}(u,v,w) = [[u,\ \ u,\ \ -\iota v,\ \ \text{INV}(w)+\iota,\ \ -w,\ \ \text{INV}(w),$$
$$-w,\ \ -\text{INV}(w),\ \ -w+\iota,\ \ \iota v+\iota,\ \ -u,\ \ -u]].$$

generates the transposition $(-u - \frac{v}{uv+\iota}, -u - \frac{v+1}{uv+u+\iota})$. This sequence is obtained by substituting the statement of Lemma B.3 into the middle part of $\gamma(u,v)$ from Lemma B.2.

Now we modify ψ'_{even} to obtain transpositions with the fixed element 0, i.e. $(0,\iota)$ for non-zero $\iota \in \mathbb{F}_n$. We do this by "relabelling" the left endpoint of the resulting transposition to be a 0. It suffices to conjugate ψ'_{even} with a permutation that maps $-u - \frac{v}{uv+\iota}$ to 0, e.g. $\text{ARK}_{u+\frac{v}{uv+\iota}}$.

Composing two ARK operations gives another ARK operation with a key equal to the sum of the two original round keys. Thus

$$\psi_{\text{even}}(u,v,w) = \text{ARK}_{u+\frac{v}{uv+\iota}} \circ \psi'_{\text{even}}(u,v,w) \circ \text{ARK}_{u+\frac{v}{uv+\iota}}$$

gives the transposition

$$\left(0, \frac{v}{uv+\iota} - \frac{v+1}{uv+u+\iota}\right) = \left(0, \frac{-\iota}{(uv+\iota)(uv+u+\iota)}\right).$$ □

C Missing Details on the Paths Constructions

C.1 Fields of even characteristic

Lemma C.1. *Let $n = 2^b$, for $b \geq 3$. Then*

$$|\{(u,v,w) \mid (0,s) = \psi_{\text{even}}(u,v,w), uv \neq 1, uv+u \neq 1, u \neq 0\}| = n(n-2).$$

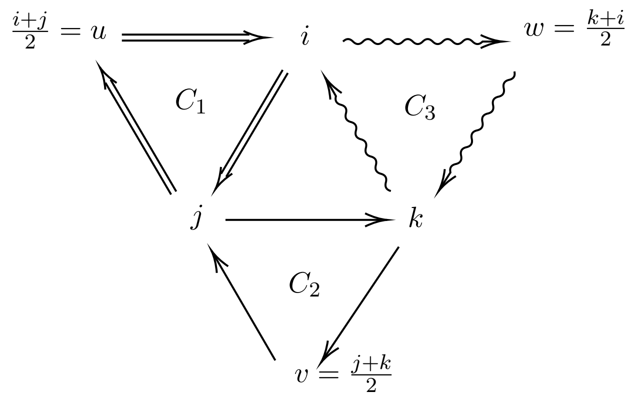

Figure 4: A schematic representation of the setting in Lemma C.2. The Claim implies that using the 3-cycles C_1, C_2, C_3 a finite number of times, we can generate the 3-cycle (i, j, k).

Proof. Recall that a valid sequence $\psi_{\text{even}}(u, v, w)$ generates the transposition $\left(0, \frac{1}{(uv+1)(uv+u+1)}\right)$. Then to generate $(0, s)$ it must hold that $\frac{1}{(uv+1)(uv+u+1)} = s$. Recalling $u \neq 0$, $s \neq 0$, we reorganize this to get

$$v(v+1) = \frac{1/s + u + 1}{u^2}.$$

Viewing this as a quadratic equation in v, it is well known that in characteristic 2, there are two solutions v if the trace from $\mathbb{F}_n$ to $\mathbb{F}_2$ of the right-hand side is equal to 0, and no solutions otherwise. By linearity of trace,

$$\mathrm{Tr}\left(\frac{1/s + u + 1}{u^2}\right) = \mathrm{Tr}\left(\frac{1/s}{u^2}\right) + \mathrm{Tr}\left(\frac{1}{u}\right) + \mathrm{Tr}\left(\frac{1}{u^2}\right) = \mathrm{Tr}\left(\frac{1}{u^2 s}\right)$$

as trace is invariant under the Frobenius homomorphism $x \mapsto x^2$. By injectivity of the Frobenius homomorphism, for a fixed s, as u varies over all nonzero values, $1/u^2 s$ varies over all nonzero values in $\mathbb{F}_n$. As $|\,\mathrm{Tr}^{-1}(0)| = n/2$, there are exactly $n/2 - 1$ nonzero values u for which $\mathrm{Tr}(1/u^2 s) = 0$.

Since each of these has two solutions v, this gives $n - 2$ pairs (u, v). As w is completely free, this gives $n(n-2)$ triples (u, v, w). $\square$

C.2 Fields of odd characteristic

Lemma C.2. *Let $n = p^b$, where p is an odd prime. Let $i, j, k \in \mathbb{F}_n$ be distinct numbers that do not form an arithmetic progression. Moreover, let $u = \frac{i+j}{2}, v = \frac{j+k}{2}, w = \frac{k+i}{2}$. Consider the following 3-cycles acting on S_n.*

$$C_1 = (i, j, u), \quad C_2 = (j, k, v), \quad C_3 = (k, i, w).$$

Then

$$C_2^2 \circ C_1 \circ C_3 \circ C_2^2 \circ C_3^2 \circ C_1 = (i, k, j).$$

Proof. Since the C_i's are 3-cycles, applying C_i twice is equal to C_i^{-1}. We will thus compute the permutation $C_2^{-1} \circ C_1 \circ C_3 \circ C_2^{-1} \circ C_3^{-1} \circ C_1$ and show that it is equal to (i, k, j). Since the 3-cycles only touch the elements i, j, k, u, v, w, it suffices to restrict our attention to these 6 elements. Furthermore, it is convenient to arrange these 6 elements in a triangle as in Figure 4. In the following proof, we also use boldface to indicate the elements that are involved in the 3-cycle that will get applied. The statement follows by a straightforward calculation:

Applying $C_2^2 \circ C_1 \circ C_3 \circ C_2^2 \circ C_3^2 \circ C_1$ gives

$$
\begin{array}{ccccc} \mathbf{u} & & \mathbf{i} & & w \\ & \mathbf{j} & & k & \\ & & v & & \end{array}
\xrightarrow{C_1}
\begin{array}{ccccc} j & & \mathbf{u} & & \mathbf{w} \\ & i & & \mathbf{k} & \\ & & v & & \end{array}
\xrightarrow{C_3^{-1}}
\begin{array}{ccccc} j & & w & & k \\ & \mathbf{i} & & \mathbf{u} & \\ & & \mathbf{v} & & \end{array}
$$

$$
\xrightarrow{C_2^{-1}}
\begin{array}{ccccc} j & & \mathbf{w} & & \mathbf{k} \\ & u & & \mathbf{v} & \\ & & i & & \end{array}
\xrightarrow{C_3}
\begin{array}{ccccc} \mathbf{j} & & \mathbf{v} & & w \\ & \mathbf{u} & & k & \\ & & i & & \end{array}
$$

$$
\xrightarrow{C_1}
\begin{array}{ccccc} u & & j & & w \\ & \mathbf{v} & & \mathbf{k} & \\ & & \mathbf{i} & & \end{array}
\xrightarrow{C_2^{-1}}
\begin{array}{ccccc} u & & j & & w \\ & k & & i & \\ & & v & & \end{array}.
$$

Applying (i,j,k) gives

$$
\begin{array}{ccccc} u & & \mathbf{i} & & w \\ & \mathbf{j} & & \mathbf{k} & \\ & & v & & \end{array}
\xrightarrow{(i,k,j)}
\begin{array}{ccccc} u & & j & & w \\ & k & & i & \\ & & v & & \end{array}.
$$

$\square$

Proof of eq. (2). We consider first the application of $\psi_{\text{odd}}(u,v)$ on some $x \notin \{-u, -u-\frac{1}{v}, -u-\frac{2}{v}\}$.

$$
\begin{aligned}
x \xrightarrow{\text{ARK}_u} x+u \xrightarrow{\text{INV}} \frac{1}{x+u} \xrightarrow{\text{ARK}_v} \frac{xv+uv+1}{x+u} \xrightarrow{\text{INV}} \frac{x+u}{xv+uv+1} \xrightarrow{\text{ARK}_{-2/v}} -\frac{xv+uv+2}{v(xv+uv+1)} \\
\xrightarrow{\text{INV}} -\frac{v(xv+uv+1)}{xv+uv+2} \xrightarrow{\text{ARK}_v} \frac{v}{xv+uv+2} \xrightarrow{\text{INV}} \frac{xv+uv+2}{v} \xrightarrow{\text{ARK}_{-u-2/v}} x.
\end{aligned}
$$

Now consider what happens when $x=-u$:

$$
-u \xrightarrow{\text{ARK}_u} 0 \xrightarrow{\text{ARK}_v \circ \text{INV}} v \xrightarrow{\text{ARK}_{-2/v} \circ \text{INV}} -\frac{1}{v} \xrightarrow{\text{ARK}_v \circ \text{INV}} 0 \xrightarrow{\text{ARK}_{-u-2/v} \circ \text{INV}} -u-\frac{2}{v}.
$$

Now consider what happens when $x=-u-\frac{1}{v}$:

$$
-u-\frac{1}{v} \xrightarrow{\text{ARK}_u} -\frac{1}{v} \xrightarrow{\text{ARK}_v \circ \text{INV}} 0 \xrightarrow{\text{ARK}_{-2/v} \circ \text{INV}} -\frac{2}{v} \xrightarrow{\text{ARK}_v \circ \text{INV}} \frac{v}{2} \xrightarrow{\text{ARK}_{-u-2/v} \circ \text{INV}} -u.
$$

Thus to complete the permutation, $-u-2/v \mapsto -u-1/v$. $\square$